

Patient Encounters & Performance Skills

Part III · PEPS

A station-by-station and item-by-item preparation manual for the twelve-station performance examination, built directly on the published blueprint and the official skills evaluation forms.

LOCATION	NCCTO, Edison Building, Charlotte, NC
STRUCTURE	12 stations: 10 encounters + 2 skills
CYCLE	3 min observation + 15 min station + 4 min break
SESSION LENGTH	4.5 hours · about 7 hours on site
EXTENDED BREAK	24 minutes after cycle 8
SCORED SKILLS	5 skills across 2 stations
RECORDING	Every cycle recorded, audio and video
SCORING	Pass or fail · no scaled score reported

Structure, timing, station content, and every scored item follow the NBEO Part III PEPS Candidate Guide, Blueprint and Content Outline, and the Anterior and Posterior Segment Skills Evaluation Forms for the August 2026 to May 2027 administration. Always verify against the current documents on the NBEO website before your exam date.

How to use this guide

Part III PEPS is unlike the two examinations before it. Nothing is multiple choice. You are scored on **what you say out loud and what you physically do**, item by item, against a published checklist, while a licensed optometrist watches and a camera records what you see through the ocular.

That has one dominant consequence for how you prepare: **knowing the answer is worth nothing unless you verbalise it, in the right words, within the fifteen minutes**. Candidates who fail rarely fail on knowledge. They fail because they examined a structure without naming it, described a finding as "normal" instead of describing it, lost a view and did not troubleshoot, ran out of time, or wrote a plan without a timeframe.

What this guide does

STRUCTURE OF THE GUIDE

Chapter	What it gives you
1. Exam architecture	The twelve stations, the timing, the roles, the vocabulary, and the day-of logistics, so nothing on the day is a surprise.
2. The patient encounter playbook	A minute-by-minute plan for the 15-minute encounter station, plus the complete focused history framework and the rules for ordering ancillary tests.
3. The encounter note	Assessment and plan templates that satisfy every documentation requirement, with worked examples.
4. The nine clinical presentations	For each blueprint category: the cases you are likely to meet, the focused history questions, the ancillary tests, the assessment, and a complete written plan.
5. Anterior segment skills station	Item-by-item scripts for biomicroscopy, 4-mirror gonioscopy, and Goldmann applanation tonometry, mapped to the official evaluation form.
6. Posterior segment skills station	Item-by-item scripts for binocular indirect ophthalmoscopy and dilated biomicroscopy with the non-contact fundus lens.
7. Findings lexicon	The exact words to use for every structure, normal and abnormal, so you never say "fine" or "within normal limits".
8. Rules, repeats, stops, and safety	Views, confirmations, attempts, repeat policy, hygiene, examiner interaction, and incident reports.
9. Rehearsal plans	Eight-week, four-week, and one-week rehearsal schedules with timed drills.
Appendix	Printable scored-item checklists for all five skills, to run mock stations with a partner.

The blueprint: how you are actually scored

COMPETENCY DOMAINS AND THEIR WEIGHTS

Competency domain	Weight	How it is assessed
Clinical Assessment and Interpretation	29	Through the electronic encounter note : your diagnosis, its location, and your interpretation of the clinical data and ancillary tests. The largest single domain.

Competency domain	Weight	How it is assessed
Management and Documentation	25	Also through the encounter note: treatment, referrals, additional testing, patient education as written, and the return-to-clinic interval.
Skills	22	Through the five physical skills in the two skills stations and through the focused case history in every patient encounter station . The case history is explicitly part of the skills sub-score.
Patient Education	13	Through what you say to the standardised patient : explaining the diagnosis, its cause, the treatment, and the follow-up in patient-centred language.
Communication and Professionalism	11	Respect, effective two-way information exchange, and collaboration with the patient and other professionals. Scored by the standardised patient against a published matrix.

CLINICAL PRESENTATION CATEGORIES: THE TOPICS OF THE TEN ENCOUNTERS

Category	Weight	Category	Weight
Anterior segment disease	17	Refraction	11
Posterior segment disease	16	Neuro-ophthalmic disease	9
Glaucoma	14	Contact lenses	8
Systemic disease	11	Binocular vision	8
		Pediatrics	6

THREE STRUCTURAL FACTS THAT SHOULD SHAPE YOUR ENTIRE PREPARATION

- Fifty-four percent of your score comes from the encounter note.** Clinical Assessment and Interpretation plus Management and Documentation total **54** of **100**, and both are captured through the electronic note. If you can examine beautifully but write a vague plan, you will struggle. Practise *writing* plans, under time pressure, in a text box.
- Skills is only twenty-two percent, and the case history is part of it.** The two skills stations matter, but they are not the majority of the exam, and a substantial part of the Skills domain is earned by taking a thorough focused history in the other ten stations. Do not spend all your preparation at a slit lamp.
- Patient Education plus Communication is twenty-four percent, and it is earned entirely by talking.** That is more than the physical skills. Rehearse explaining ten diagnoses out loud, in plain language, to a non-clinician.

How to rehearse, and why rehearsal beats reading here

1. **Say everything out loud, always.** From the first week of preparation, never practise a skill silently. The examiner cannot award an item for a finding you noticed but did not state, and the habit of narrating has to be automatic before you are under pressure.
2. **Rehearse against the actual checklist.** The evaluation forms are published. The appendix reproduces them as checklists. Have a classmate score you item by item, aloud, and count the misses.
3. **Always practise on a clock.** Every station is exactly **15 minutes**. Set a timer for every drill from the beginning. Time is the single commonest reason candidates lose items they knew.
4. **Practise on unfamiliar people.** The standardised patients are strangers, they are trained to be neutral, they may blink or ask for a break, and they may be portraying a person quite unlike themselves. Practise on people you have not examined before.
5. **Rehearse the note in a text box, not on paper.** You will type the assessment and plan on a laptop while a person sits in front of you. Practise typing a complete plan in under **4 minutes**.
6. **Rehearse the patient explanation on a non-clinician.** If your partner or flatmate cannot restate what you told them, the standardised patient will not be able to either.
7. **Build a fixed sequence for each skill and never vary it.** Under stress you will fall back on habit. Make the habit the checklist order.

THE SIX FAILURE MODES, RANKED

1. **Silence.** Examining a structure without naming it and describing it.
2. **Vague description.** Saying "normal", "fine", "OK", "unremarkable", or "within normal limits" instead of describing what you see.
3. **Running out of time,** usually because of an over-long history or a lost view that was not troubleshooted quickly.
4. **An incomplete plan.** No dose, no duration, no referral timeframe, or no return-to-clinic interval.
5. **Losing or never obtaining the view,** and not asking whether the examiner has a view.
6. **Hygiene and safety breaks,** most often touching the face or hair after washing hands, or failing to state that instruments would be disinfected.

Every one of these is a rehearsal problem rather than a knowledge problem, and every one is fixable in the weeks before the exam.

A note on currency. NBEO updates the candidate guide, the blueprint, and the evaluation forms annually. This guide reflects the August 2026 to May 2027 documents. Download the current versions from the NBEO website when you register and reconcile them against the checklists in the appendix, because item numbering and wording do change between administrations.

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CHAPTER 1

Exam architecture, vocabulary, and day-of logistics

12 stations · 4.5 hour session

NCCTO, Charlotte NC

All cycles audio and video recorded

KNOW THESE COLD BEFORE YOU TRAVEL

1. The station structure, the cycle timing, and where the extended break falls.
2. Who is in each room, and what each of them will and will not tell you.
3. The three announcements and exactly what each one signals.
4. What you may and may not do during observation time.
5. What identification and attire are required, and what may not enter the building.
6. How and when to file an incident report.

CYCLE LENGTH

15

minutes per station,
preceded by 3 minutes of
observation and followed by
a 4 minute break.

The twelve stations

ONE SESSION = 12 STATIONS, in a randomly assigned rotation order, within one assigned pod (blue or green)

10 x PATIENT ENCOUNTER STATIONS	2 x PERFORMANCE SKILLS
-----	-----
Focused case history	ANTERIOR SEGMENT STATION
Review clinical data on the device	1. Biomicroscopy
Order ancillary tests (final on submit)	2. 4-mirror gonioscopy
Interpret results	3. Goldmann applanation
Verbal patient education	POSTERIOR SEGMENT STATION
Type the encounter note (A and P)	4. Binocular indirect
9 of 10 are scored.	ophthalmoscopy
1 of 10 is a data-collection station	5. Dilated biomicroscopy
(unidentified, does not count).	+ non-contact fundus
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EACH CYCLE: 3 min observation -> 15 min scored station -> 4 min break
AFTER CYCLE 8: extended break of 24 minutes

THE TIMING YOU SHOULD INTERNALISE

Element	Duration	Notes
Observation time	3 minutes	Before every station. Overview information appears on your device.
Exam cycle	15 minutes	All stations, encounter and skills alike. Only actions inside this window are scored.
Break	4 minutes	In the hallway at your designated desk for the <i>next</i> room number.
Extended break	24 minutes	After the eighth cycle. Snacks stored in hallway bins are accessible during breaks.
Whole session	~4.5 hours	Plus check-in, orientation, and check-out: expect about 7 hours on site.

THE THREE ANNOUNCEMENTS, AND WHAT EACH MEANS FOR YOU

Announcement	Action
"Candidates, please enter the exam room."	Enter, close the door behind you , and begin your 3 minutes of observation. The pod colour is stated in the announcement. In a skills station the standardised patient is already in the room; in an encounter station they are not, and will enter shortly before the cycle begins.
"Exam cycle [number] has begun."	The scored 15 minutes starts. In encounter stations, the clinical data becomes available on your device at this moment. You may now start asking the standardised patient questions and, in skills stations, you may now touch the patient.
"Exam cycle [number] has ended, please exit the room."	Stop, leave the room, and go to your designated break-area desk for the next room number. The encounter note submits automatically. Remain in the room until this plays , even if you have finished.

Who is in the room

ROLES, AND WHAT EACH WILL TELL YOU

Role	Present in	What they will and will not do
Standardised patient (SP)	Every station	A trained individual portraying the case. All SPs share the fictitious family name Lee , so "Mr. Lee" or "Ms. Lee" is always correct. They answer history questions only from their script ; anything not on the script gets "I do not have that information" . They will not initiate casual conversation beyond a cordial greeting. They score Communication and Professionalism. In a skills station they remain for the whole cycle; in an encounter station they leave only if you dismiss them.
Skills examiner	Skills stations	A licensed optometrist scoring your performance item by item. Will confirm the <i>presence</i> of a view if asked, may give instructions, may intervene for safety, and will stop you after four unsuccessful attempts at an item.
Proctor	Posterior segment skills station	NCCTO staff who can confirm views. Same rules as the examiner regarding views.
Encounter examiner	Not in the room	A licensed optometrist who scores your submitted encounter note afterwards. Only the SP is physically present in an encounter station.
NCCTO staff and hallway monitor	Hallway	Administer the exam, answer "where" questions, escort to the restroom, and resolve equipment problems.

Role	Present in	What they will and will not do
Observers	Occasionally any station	NBEO-approved personnel. They do not affect your score, will not speak to you, and should be ignored.

WHAT YOU CAN AND CANNOT ASK

- **You may ask "where" questions** at any time, during observation and during the cycle: where the lighting control is, where a supply is kept.
- **You may ask whether the examiner or proctor has a view**, at any point while you are performing that view.
- **You may not ask how to use equipment** or any other instructional question. It will not be answered.
- **Confirmation of a view means only that something is visible.** It does not tell you whether the view is good enough, whether it meets the item criteria, or whether it is of the correct structure or quadrant.
- **Examiners comment on a view only while it is being performed.** Asking afterwards gets "I can only comment on a view while it is being performed."
- **Neutral responses are not feedback.** "I cannot answer that", "I do not have that information", and "it is up to you" are scripted and tell you nothing about your performance.
- **No time is added** for asking and answering questions.

Observation time: use all three minutes

WHAT YOU MAY AND MAY NOT DO DURING OBSERVATION

In a skills station you may	You may not
• Read the station overview information on the device • Wash your hands • Disinfect the slit lamp with an alcohol wipe • Focus the slit lamp and set the illumination • Try on the BIO headset and adjust pantoscopic tilt and the oculars • Familiarise yourself with slit lamp controls and room lighting • Set out supplies (without opening packages, except alcohol wipes) • Read the repeat policy posted on the door • Ask "where" questions of the SP or hallway monitor	• Perform any action on the standardised patient, including asking them to move into the slit lamp • Open any sealed package other than an alcohol wipe In an encounter station the SP is not present during observation. The overview information is limited to patient demographics, the chief complaint, the review of systems, and a current medication list, and it remains available as part of the clinical data once the cycle starts. You must at minimum use hand sanitiser before entering; hand washing is not required in encounter stations.

A THREE-MINUTE OBSERVATION ROUTINE, MEMORISED**SKILLS STATION**

- 1 **0:00 to 0:20.** Read the overview: which skills, in this station. Read the repeat policy on the door.
- 2 **0:20 to 0:50. Wash your hands** to CDC standard. Then **do not touch your face or hair.**
- 3 **0:50 to 1:20.** Disinfect the slit lamp headrest and chin rest with an alcohol wipe. Locate the anaesthetic, fluorescein, cushioning solution, cotton-tipped applicators, tissues, and waste bin.
- 4 **1:20 to 2:20. Focus the slit lamp on the headrest bar,** set your oculars to your own refraction, set the interpupillary distance, and set the beam: low magnification, wide diffuse beam to start, and check that you can produce a parallelepiped and an optic section quickly.
- 5 **2:20 to 2:50.** If this is the posterior station, fit the BIO headset, set the pantoscopic tilt and the oculars, and confirm you can see clearly at **18 to 25 inches**. Locate the condensing lens and the non-contact fundus lens. Practise the room lighting control.
- 6 **2:50 to 3:00.** Rehearse your opening sentence silently, and decide which skill you will start with.

ENCOUNTER STATION

- 1 **0:00 to 0:30.** Sanitise your hands. Read the demographics and the chief complaint. Say the likely differential to yourself.
- 2 **0:30 to 1:30.** Read the review of systems and the medication list carefully. **Every medication is a clue:** it is the cause, a contraindication, or a diagnostic pointer.
- 3 **1:30 to 2:30.** On the note board, write your history plan: the three or four HPI questions that will most efficiently separate your differential, plus the standing prompts (ocular, medical, medication, allergy, surgical, family, social).
- 4 **2:30 to 3:00.** Rehearse your opening: introduce yourself by OE TRACKER number, confirm the patient, and state what you are going to do.

Vocabulary from the candidate guide

TERMS YOU WILL HEAR USED PRECISELY

Term	Meaning
Exam cycle	The timed 15 minutes of a single station. Only actions inside it are scored.
Observation time	The 3 minutes before the cycle begins.
Overview information	What appears on your device during observation: in encounters, demographics, chief complaint, review of systems, and medications; in skills stations, a reminder of the skills to be performed.
Device	The NBEO-provided laptop running the exam platform.
Note board	An electronic writing tablet in the encounter rooms. It stays in the room and is not scored .
Item	A single numbered scored element on the evaluation form.
Finished	The state after you declare you are finished with a skills station; you are no longer scored on further items.
Repeat	Redoing an item or a whole skill. Announce a skill repeat before starting it.

Term	Meaning
Stop	When your performance is halted for safety or after four unsuccessful attempts at an item. Remaining items in that skill are scored "no".
Intervene	When the SP, proctor, or examiner interrupts. You are expected to take corrective action before proceeding.
Incident report	An optional report filed on your device during the break after a station, documenting any irregularity you think affected your performance.
Session	A complete 12-station examination.
Pod	One of two sets of 12 stations, blue or green. You remain in your assigned pod, and its colour is named in every announcement.

Day-of logistics

REQUIREMENTS AND PROHIBITIONS

Item	Requirement
Reporting	NCCTO, Edison Building, 2nd floor, 7910 Microsoft Way, Charlotte, NC 28273, at the date and time in your registration email. Arriving on the wrong date or time cannot be accommodated, and late check-in can mean disqualification with forfeiture of the fee. Arrive no earlier than 15 minutes before your listed time; entry is not permitted before then. No guests.
Identification	One valid government-issued photo ID to enter the building, then two valid forms of personal identification inside , both signed and unexpired, at least one government-issued with a photo, and both matching your registration name.
Account details	You must know your OE TRACKER number and your NBEO password to check in and to sign into the device.
Attire	Professional attire, and you must bring and wear a white lab coat throughout . Excluded: jeans, shorts, athletic pants, T-shirts, scrubs, immodest garments, tennis shoes, sneakers, and flip-flops. Collared short-sleeved shirts are acceptable. Inappropriate attire may prevent you from testing.
Armbands	Two armbands with your candidate badge number, worn throughout, with the number displayed on the side of the arm . The reverse carries your name and OE TRACKER number for verification. Introduce yourself by your OE TRACKER number at the start of each cycle.
Photograph	Taken at check-in. Appear as you will during the exam: wear your glasses if you plan to, tie your hair back if you plan to.
Equipment	All equipment is provided. Do not bring your own. Review the Site Information and Equipment List in advance so you know the makes and models you will use.
Lockers and personal items	Lockers are provided (16.5 deep by 10.5 wide by 27.5 inches tall). Cell phones, watches, wallets, purses, and backpacks go in the locker. Luggage is not permitted in the building. No firearms or weapons of any kind.
Notes and materials	No notes, written or electronic, may enter the orientation room or the testing area at any time. Anything found may be treated as improper conduct.

Item	Requirement
Watches	Not allowed in the exam rooms. Use the countdown timer in the platform, which displays the official time. Examiners, proctors, and SPs will not remind you of remaining time.
Reasonable items permitted	Tissues, cough drops, mints, gum, lip balm, hair ties and clips, sanitary items, and an eye patch if you use one. All items must be shown to staff for approval.
Items that stay outside the room	Drinks (labels removed, lid on) and individually wrapped snacks, stored in hallway bins and accessible during breaks. There is no refrigerator.
Restroom and water	Use the restroom before starting. No extra time is given. During the exam, tell the SP or hallway monitor and you will be escorted; otherwise use break time.
Candidate-to-candidate contact	Permitted only during check-in and orientation. Once escorted to the exam hallway, no communication with other candidates of any kind, including gestures, expressions, thumbs up or down, and passing notes, until you have finished and left the NCCTO.
Recording	Every cycle is recorded, audio and video. Assume any conversation, including a post-exam discussion with staff in a room, is recorded.
Wounds and injuries	Open wounds on a finger or hand must be covered; show any wound to staff at check-in. If you are injured during the exam, notify staff immediately, use universal precautions, and stop any bleeding before continuing. No extra time is given for an injury caused by candidate error.
Monocular or amblyopic candidates	Use your better eye at the left ocular of the slit lamp, since the camera is mounted there. This may mean repositioning at the slit lamp or wearing a patch over the right eye. You must bring your own patch, and this does not require a formal accommodation.
Test accommodations	Requested in advance through the NBEO Test Accommodations Policy, under the Americans with Disabilities Act.

INTERACTING WITH THE STANDARDISED PATIENT

- Address them as **Mr. Lee or Ms. Lee**. Introduce yourself by your **OE TRACKER number**.
- **Handshakes are not permitted**, for hygiene.
- **Do not touch the patient below the shoulder**. Be mindful of personal space, especially at the slit lamp.
- Respond to **the patient being portrayed**, not to the SP's real appearance. A forty-year-old woman may be portraying an elderly man who wears glasses. If the case is a child, the SP will tell you they are the parent or guardian.
- Casual conversation may occur **only if you initiate it**, and there is a list of topics that must not be discussed: NBEO or NCCTO, the exam, the SP's job, your performance, and your school.
- If you need to read the device for a while, you may say **"Please give me a moment to review some information on my device."** This is not required, and you are not penalised for using the device, but it eases the silence and the SPs are trained to expect pauses.
- To dismiss the SP in an encounter station, say something explicit such as **"Thank you, Ms. Lee, you may check out."** **"Thank you" alone will not do it.** Once they leave they cannot return, so only dismiss them when you are certain you have finished all questions and all patient education.
- In a skills station, if you want to end scoring early, say **"I am finished with this station."** The examiner then leaves and no further items are scored. This is your choice, and there is rarely a reason to do it.
- **Stay in the room until the ending announcement plays** regardless.

Incident reports and irregularities

- File on your device **during the break immediately after the affected station**. Reports are not accepted once you have left the exam hallway.
- Document anything you believe negatively affected your performance: equipment, technology, the examiner, proctor, or SP, or a problem with your own performance caused by an irregularity.
- **Spelling and grammar do not matter**. Content does. Be specific about what happened, when, and how it affected you.
- All reports are reviewed. Staff may review the recording or interview the SP or examiner, and may speak with you after the session. If they do so in an exam room, assume it is recorded.
- **A redo may be offered for a skills station only**, and only for an administrative irregularity such as equipment failure, power loss, or an SP problem, limited to the affected skill or station, and only on the day of your exam. **A redo is never offered for a patient encounter station.**
- During the cycle, if equipment fails, **tell the SP**, who will summon staff. If extra time is granted it appears in the top left corner of your device.

RETRIEVAL PRACTICE

1. How many stations are there, how are they split, and how many of the encounters are scored?
2. Give the three timing intervals in a cycle and where the extended break falls.
3. What are you permitted to do to the standardised patient during observation time?
4. Name three things you should do during observation in a skills station that save time later.
5. What does the examiner confirming a view actually tell you, and what does it not tell you?
6. What will the SP say if you ask a history question not on their script?

7. How do you introduce yourself, and how do you address the patient?
8. What exact phrase dismisses the SP from an encounter station, and why does "thank you" not work?
9. Where is the slit lamp camera mounted, and what does that mean for a monocular candidate?
10. When can you file an incident report, and for which station type can a redo be offered?
11. What identification do you need to enter the building versus to enter the testing area?
12. Name four items that must go in your locker and two that are permitted in the room.
13. How do you keep track of time?
14. What are you forbidden from doing with other candidates once you are in the exam hallway?

KEY

1. Twelve stations: **10 patient encounters and 2 skills stations. 9 of the 10 encounters are scored**; one is an unidentified data-collection station.
2. **3 minutes** observation, **15 minutes** station, **4 minutes** break; the **24 minute** extended break falls **after cycle 8**.
3. **Nothing**. You may not perform any action on the SP, including asking them to move into the slit lamp.
4. Wash your hands, disinfect the slit lamp, and focus the slit lamp and set the oculars (also fit the BIO headset and set out supplies without opening them).
5. It tells you only that **something is visible**. It does not tell you the quality of the view, whether it meets the item criteria, or whether it is the correct structure or quadrant.
6. **"I do not have that information."**
7. Introduce yourself by your **OE TRACKER number**; address the patient as **Mr. or Ms. Lee**. No handshake.
8. **"Thank you, Ms. Lee, you may check out"** or similar explicit dismissal; "thank you" alone is not a signal to leave.
9. On the **left ocular**. A monocular or amblyopic candidate should use their better eye at the left ocular, repositioning or patching the right eye, and must bring their own patch. No formal accommodation is needed for this.
10. During the **break immediately after the affected station**, and never after leaving the exam hallway. A redo may be offered for a **skills station only**, for administrative irregularities, on the day.
11. **One government-issued photo ID** to enter the building; **two valid signed, unexpired forms** inside, at least one government-issued with a photo.
12. Locker: cell phone, watch, wallet or purse, backpack, and all notes and study materials. Permitted in the room: tissues, cough drops or mints or gum, lip balm, hair ties, sanitary items, and an eye patch.
13. The **countdown timer in the exam platform**, which shows the official time. Watches are prohibited and nobody will remind you.
14. Any communication at all, including verbal, gestures, facial expressions, thumbs up or down, and passing notes, until you have completed the exam and left the NCCTO.

CHAPTER RECAP

1. Twelve stations, ten encounters and two skills stations, 15 minutes each, with a 24 minute break after cycle 8.
2. Three minutes of observation is free preparation time: wash, disinfect, focus, and plan.
3. You may not touch the patient during observation, and you may not open packages.
4. Only "where" questions and view confirmations are answered; a confirmed view means only that something is visible.
5. Introduce yourself by OE TRACKER number; address the patient as Mr. or Ms. Lee, no handshake, nothing below the shoulder.
6. Dismiss the SP explicitly, and only when you are certain you are finished.
7. Use the platform timer; there are no watches and no reminders.
8. The slit lamp camera is on the left ocular, so what you see through the left ocular is what is scored.
9. File incident reports in the break immediately after the affected station.
10. No communication with other candidates once you are in the hallway, and everything is recorded.

CHAPTER 2

The patient encounter station playbook

10 of 12 stations

Case history counts toward the Skills domain

Patient education 13% • Communication 11%

ANSWER THESE COLD BEFORE READING

1. Give a minute-by-minute plan for the 15-minute encounter station.
2. List the six categories of history you must cover and the HPI elements within the first.
3. State the rules for ordering ancillary tests, including what happens on submission.
4. State what the clinical data will contain and when it becomes available.
5. Give the structure of a patient education explanation in patient-centred language.
6. State how you dismiss the SP and when you should not.

DOMAINS EARNED HERE

78%

Assessment and interpretation, management and documentation, patient education, communication, and the history component of skills.

The fifteen minutes, minute by minute

0:00 ENTER SCORED TIME. Introduce (OE TRACKER number), confirm patient, state the plan. Clinical data becomes available on the device at this moment.

0:30 FOCUSED CASE HISTORY. Chief complaint HPI first, then the standing categories. Talk to the patient, not the screen. Take brief notes on the note board.

4:30 REVIEW CLINICAL DATA on the device. Say "Please give me a moment to review some information on my device." Read pretest, refraction, anterior, posterior.

6:30 ASK ANY FOLLOW-UP HISTORY QUESTIONS the data has raised. This is allowed and often the highest-yield 60 seconds in the station.

7:30 ORDER ANCILLARY TESTS with indications. SUBMISSION IS FINAL.

8:30 REVIEW ANCILLARY RESULTS. Interpret. Commit to a diagnosis.

9:30 PATIENT EDUCATION, verbal, in plain language: what it is, why it happened, what we will do, what to watch for, when to come back.

11:00 DISMISS THE SP if you are certain you are finished with them.
"Thank you, Ms. Lee, you may check out."

11:00 TYPE THE ENCOUNTER NOTE: Assessment (from the searchable list, with location) and Plan (treatment, referrals, additional testing, patient education, RTC).

14:30 REREAD the plan. Check: dose, duration, referral specificity, RTC interval.

15:00 CYCLE ENDS. The note submits automatically.

THREE TIMING RULES THAT DECIDE THE STATION

1. **Cap the history at four minutes.** A thorough focused history is scored; an exhaustive one costs you the note. Four minutes is enough for a complete focused history if you have rehearsed the sequence.
2. **Start the note by minute 11 at the latest.** The note carries 54% of the blueprint. Leaving under four minutes for it is the single most common self-inflicted failure.
3. **Do the patient education before you type.** If you type first and run out of time, you lose Patient Education and Communication, which together are 24%. If you educate first and run short on typing, you can still capture the essentials in bullet form.

The focused case history

The case history is **explicitly part of the Skills sub-score**, and it is scored on **what you verbalise**. You do not need to enter subjective information into the note. That means you must actually ask the questions out loud, not simply infer the answers.

THE SIX CATEGORIES, AND WHAT TO ASK IN EACH

Category	Ask
1. History of the present illness	Location ("Which eye, or both?"), onset ("When did it start? Was it sudden or gradual?"), duration ("How long has it lasted? Is it constant or does it come and go?"), severity ("On a scale of one to ten?" or "How much is it affecting what you do?"), frequency ("How often?"), modifying factors ("What makes it better or worse?"), associated symptoms ("Anything else with it: pain, redness, discharge, flashes, floaters, headache, nausea?"), prior episodes ("Has this happened before? What was done?"), and further description ("Describe it in your own words.").
2. Past ocular history	"Have you ever been diagnosed with any eye condition?" plus, for each: how long, and what treatment. Then: eye surgery or laser , eye injury , date of your last eye examination , do you wear glasses or contact lenses , what type, how old is the prescription , and how compliant are you with wear and replacement . For contact lens cases: wear schedule, overnight wear, solution, water exposure, replacement interval, and comfort.
3. Medical history and medications	"Do you have any medical conditions?" and for each: how long and what treatment and how well controlled. Date of last physical examination . All medications including over-the-counter, supplements, eye drops, inhalers, injections, and creams . Allergies, and what reaction . Then relevant specifics: for a diabetic, the most recent HgbA1c; for a hypertensive, the most recent blood pressure.
4. Surgical history	"Any surgeries or procedures, on the eyes or elsewhere?" and when.
5. Family history	Family eye history : glaucoma, macular degeneration, retinal detachment, blindness, strabismus or lazy eye, keratoconus, and any inherited eye disease. Family medical history : diabetes, hypertension, heart disease, stroke, autoimmune disease, and cancer.
6. Social history	Occupation and the visual demands of it. Hobbies . Driving . Smoking, alcohol, and recreational drug use . Risky behaviours where relevant, including sexual history when the differential includes a sexually transmitted infection. Eye protection use at work and in sport. Living situation and support, where relevant to management.

A REHEARSED OPENING AND HISTORY SEQUENCE

"Good morning, Ms. Lee. My name is candidate [OE TRACKER number] and I am the optometrist seeing you today. I am going to ask you some questions about your eyes and your health, then review your test results, and then we will talk about what I find and what we will do. Is that all right?"

"Tell me in your own words what brought you in today."

"Which eye is it, or is it both? When did it start? Was it sudden or gradual? Is it constant or does it come and go? On a scale of one to ten, how bad is it? Does anything make it better or worse? Have you noticed anything else along with it, such as pain, redness, discharge, flashing lights, floaters, or headache? Has this ever happened before?"

"Have you ever been told you have any eye condition? Have you had any eye surgery, laser, or injury? When was your last eye examination? Do you wear glasses or contact lenses, and how old is that prescription?"

"Do you have any medical conditions? When was your last physical? What medications are you taking, including anything over the counter, supplements, and any eye drops? Do you have any allergies, and what happens when you take them?"

"Have you had any surgeries or procedures? Does anyone in your family have eye problems such as glaucoma, macular degeneration, or a retinal detachment? Any family history of diabetes, high blood pressure, or autoimmune disease?"

"What do you do for work, and what are your eyes doing all day? Do you drive? Any hobbies? Do you smoke, or drink alcohol? Do you wear eye protection at work or for sport?"

HISTORY MISTAKES THAT COST ITEMS

- **Asking closed compound questions.** "Any pain, redness, or discharge?" often gets a single answer covering only one element. Ask them separately when the differential depends on them.
- **Skipping medication and allergy questions** because the list is already on the device. The list on the device is data; **asking is a scored behaviour**, and it also lets you ask about compliance, over-the-counter agents, and drops that may not be listed.
- **Not asking about compliance and prescription age** in a refractive or contact lens case.
- **Omitting social history** because it feels irrelevant. Occupation, driving, smoking, and eye protection are scored and frequently change the management.
- **Failing to ask follow-up questions after seeing the data.** You may question the SP at any time in the cycle. Once you have the clinical data, one or two targeted questions often clinch the diagnosis, and this is a strong candidate behaviour.
- **Talking to the screen instead of the patient.** Communication is scored by the SP.

Clinical data on the device

- Becomes available **the moment the cycle begins** and remains until the cycle ends.
- Contains, where relevant to the case: **pretest information, refractive information, anterior segment evaluation, and posterior segment evaluation**, plus the overview information you already read.
- You may move between tabs freely: clinical data, ancillary tests, and assessment and plan.
- You may review data, ask questions, order tests, and write the note **in any order**.
- **Read it actively.** Some findings will be normal and some abnormal, and it is your job to identify which. Numbers are given without interpretation.

HOW TO READ THE CLINICAL DATA IN NINETY SECONDS

- 1 **Acuity and pupils first.** Reduced acuity plus an afferent defect changes the whole differential toward optic nerve or extensive retinal disease.
- 2 **IOP next.** A number outside the teens reframes the case immediately.
- 3 **Then the section that matches the chief complaint,** in detail.
- 4 **Then everything else, scanning for the one abnormal line.** Cases are constructed with a small number of deliberate abnormal findings; find them all before committing.
- 5 **Then ask yourself the negative question:** what is conspicuously *absent* that would have supported my leading diagnosis? Absent findings are as informative as present ones.
- 6 **Finally, reconcile the data with the history.** If they do not fit, you have the wrong diagnosis, or there is a second one.

Ordering ancillary tests

THE RULES, FROM THE CANDIDATE GUIDE

Rule	Detail
You type the tests	There is no menu. Type the test name and an indication for each.
Be as specific as you would be clinically	"Macular OCT" not "OCT". "Westergren ESR" not "inflammatory markers". "CN V assessment" for corneal sensation.
Grouped panels are acceptable when they are clinically ordered as a group	A CBC or a metabolic panel may be ordered as such. Anything normally ordered individually, such as a specific antibody or antigen test, must be listed separately . If unsure, list separately.
Give an indication	Each test needs a reason: "Westergren ESR, indication: rule out giant cell arteritis"; "Macular OCT, indication: evaluate for choroidal neovascular membrane"; "Contact lens fitting and assessment, indication: determine whether contact lenses will improve acuity".
Assume full scope and full access	You may order any test, laboratory investigation, or imaging study, on the understanding that you practise at full scope and the patient has access to it.
Submission is final	Once you select "Submit ancillary tests" you cannot add any more in that cycle. Write the full list before submitting.
You may receive results you did not order	The results provided are those associated with the case and may not match your list. Receiving an unordered result does not mean you lost points: you are scored on what you ordered, and multiple test sets can be correct. Some results will be normal, and recognising that is part of the task.
Results remain available	Until the cycle ends.

A DEFAULT ANCILLARY TEST LIST BY PRESENTATION

Presentation	Order, with indications
Reduced central vision with distortion	Macular OCT (evaluate for CNV, fluid, or traction); fundus photography (document); fluorescein angiography (characterise a neovascular membrane); OCT angiography; Amsler grid.
Suspected glaucoma	Central corneal thickness pachymetry; threshold visual field 24-2 (and 10-2 if central involvement); OCT retinal nerve fibre layer and ganglion cell analysis; gonioscopy; optic disc photography; diurnal IOP curve.
Optic neuropathy, any	Threshold visual field; OCT RNFL; colour vision (HRR); MRI brain and orbits with gadolinium and fat suppression; and over age 50 Westergren ESR, CRP, and platelet count to rule out giant cell arteritis.
Retinal vascular occlusion	Carotid duplex ultrasonography; ECG with rhythm monitoring; echocardiography; fasting glucose and HgbA1c; lipid panel; blood pressure; and in a young patient a thrombophilia screen (antiphospholipid antibodies, homocysteine, protein C and S, factor V Leiden).
Uveitis, granulomatous or recurrent	ACE and lysozyme; chest radiograph or chest CT; RPR and treponemal-specific antibody; interferon-gamma release assay; HLA-B27 where the pattern fits; ANA in a child.
Corneal ulcer	Corneal scraping for Gram stain and culture (blood, chocolate, Sabouraud, and non-nutrient agar with <i>E. coli</i> overlay); culture of the contact lens, case, and solution; corneal sensation (CN V assessment); anterior segment photography.
Diabetic retinopathy	Macular OCT; fundus photography; HgbA1c; blood pressure; lipid panel; urine albumin-to-creatinine ratio; fluorescein angiography or OCT angiography if ischaemia is in question.
Refractive or binocular vision complaint	Cycloplegic refraction; near point of convergence; AC/A determination; accommodative amplitude, facility, and lag (MEM); vergence ranges; stereoacuity; and a trial frame or contact lens over-refraction where relevant.
Contact lens problem	Corneal topography; corneal staining assessment with fluorescein; contact lens fitting and assessment; culture if infiltrative; specular microscopy if endothelial concern.
Pediatric case	Cycloplegic refraction; cover test at distance and near; stereoacuity; colour vision; dilated fundus examination; and, where the presentation warrants, B-scan ultrasonography or MRI.
Systemic disease presentation	Blood pressure; fasting glucose and HgbA1c; lipid panel; CBC; comprehensive metabolic panel; ESR and CRP; and disease-specific serology.

Patient education, spoken

FIVE THINGS TO SAY, IN THIS ORDER

- 1 **What it is.** Name the diagnosis (the one place medical terminology is expected), then immediately translate it. "You have what we call open-angle glaucoma. That means the drainage system inside your eye is not working efficiently, so the pressure is a little high and it is slowly damaging the nerve that carries the picture from your eye to your brain."
- 2 **Why it happened.** The cause or mechanism, in plain language and without blame. "We do not know exactly why it starts, but it runs in families and it becomes commoner with age."
- 3 **What we are going to do.** The treatment, with how and when. "I am going to start you on a drop called latanoprost. One drop in each eye every night at bedtime. It works by helping fluid drain out of the eye. Your eyes may look a little red at first and your lashes may grow longer over time."
- 4 **What to watch for.** Warning signs that mean come back sooner. "If you notice pain, redness that does not settle, or a sudden change in vision, I need to see you straight away."
- 5 **When to come back.** A specific interval. "I want to see you again in six weeks to check that the pressure has come down."

Then check understanding: "Just so I know I have explained it clearly, can you tell me how you are going to use the drop?" This is worth doing every time: it is good communication, it is scored, and it catches misunderstanding.

Language rules. Name the diagnosis, then drop the jargon entirely. Do not say "intraocular pressure", "neovascularisation", "posterior vitreous detachment", "photocoagulation", or "topical". Say pressure inside the eye, new abnormal blood vessels, the jelly inside the eye pulling away, laser treatment, and eye drops. Avoid frightening absolutes, but **do not soften a serious diagnosis into meaninglessness either**: the patient needs to understand why the follow-up matters.

ADDITIONAL EDUCATION ELEMENTS TO INCLUDE WHEN THE CASE CALLS FOR THEM

Element	Example phrasing
Home care	"Warm compresses for five minutes twice a day, then massage the lid margins like this."
Activity or lifestyle change	"Stop sleeping in your lenses, and never rinse them in tap water." "Stopping smoking will do more for this than anything I can prescribe."
Alternative options	"There is also a laser treatment that works about as well as drops for many people, if you would prefer not to use a drop every day."
Side effects	"This drop can sting for a moment and can make your eyes look red. It can also darken the colour of the iris over years, usually only in one eye if we only treat one."
Prognosis	"With treatment, most people keep the vision they have. What we cannot do is get back what has already been lost, which is why catching it now matters."
Urgency	"This is urgent. I am arranging for you to be seen by a retinal surgeon today, and you should not eat or drink until they have seen you."
Referral explanation	"I am sending you to a rheumatologist, because the inflammation in your eye is a sign of a condition affecting your whole body that needs treatment beyond eye drops."
Driving and occupation	"With the field loss in both eyes, you should not drive until you have had a formal assessment. I know that is difficult, and I will help you arrange it."

Element	Example phrasing
Family screening	"Because this runs in families, your brothers, sisters, and children should have their eye pressure and nerves checked."

Common encounter station errors

WHAT GOES WRONG, AND THE FIX

Error	Fix
History runs to eight minutes	Rehearse the six-category sequence until it takes four minutes. Time it every single practice run.
Submitting ancillary tests too early	Write the full list in the box first, reread it, then submit once.
Ordering vaguely	Name the exact test and the exact indication. "OCT" is not a test; "macular OCT to evaluate for subretinal fluid" is.
Educating with jargon	Name the diagnosis, then use only words a non-clinician uses.
Dismissing the SP too early	Only dismiss after you have finished all questions <i>and</i> all education. They cannot come back.
Starting the note with three minutes left	Put a hard mental checkpoint at minute 11.
Not reconciling data with history	Ask the negative question: what is missing that should be there?
Silence while reading the device	Say "Please give me a moment to review some information on my device", then work.
Talking to the screen	Turn your body toward the patient when you speak to them.
Forgetting the paediatric proxy	If the SP says they are a parent, direct the history and the education to them as the parent of the child described.

RETRIEVAL PRACTICE

1. Give your minute-by-minute plan, and state the two hard checkpoints.
2. Name the six categories of history and four HPI elements.
3. Is subjective information entered into the note? How is the history scored?
4. What happens when you submit ancillary tests?
5. You receive a test result you did not order. What does that mean?
6. Can you order a CBC as a single item? Can you order "antibody testing" as a single item?
7. When does clinical data become available, and how long does it remain?
8. May you ask the patient more questions after reading the clinical data?
9. Give the five components of a spoken patient education, in order.
10. Which single sentence would you add at the end of the education, and why?
11. What phrase eases a long pause while you read the device?
12. When should you dismiss the SP, and what must be true first?
13. Why do patient education before typing the note?
14. Give the ancillary test list you would order for a 74-year-old with sudden altitudinal field loss and a swollen disc.

KEY

1. Introduce and history to 4:30, data to 6:30, follow-up questions to 7:30, order tests, review results, educate from 9:30, dismiss and type from 11:00, reread at 14:30. Hard checkpoints: **history done by 4:30 and note started by 11:00**.
2. HPI, past ocular history, medical history and medications, surgical history, family history, and social history. HPI elements include location, onset, duration, severity, frequency, modifying factors, associated symptoms, and prior episodes.
3. **No**. Subjective and objective information is not documented by the candidate; the case history is scored on **what you verbalise** to the SP.
4. **Submission is final**; you cannot add further tests during that cycle.
5. Nothing about your score. Results provided are those associated with the case and may not match your list; you are scored on what you ordered, and multiple test sets may be correct.
6. Yes, a CBC may be ordered as a commonly grouped panel. **No**, antibody or antigen testing must be listed as specific individual tests.
7. At the moment the cycle begins, and it remains until the cycle ends.
8. **Yes**, at any time during the cycle, and doing so is a strong candidate behaviour.
9. What it is, why it happened, what we will do, what to watch for, and when to come back.
10. A check of understanding: "Can you tell me how you are going to use the drop?" It is good communication, it is scored, and it catches misunderstanding.
11. "Please give me a moment to review some information on my device."
12. Only when you have completed all history questions and all patient education, because once they leave they cannot return.
13. Because Patient Education and Communication together are **24%** of the blueprint and can only be earned while the patient is present, whereas a partially completed note still captures the key elements.
14. Westergren ESR, CRP, and platelet count to rule out giant cell arteritis; threshold visual field; OCT retinal nerve fibre layer; colour vision; optic disc photography; blood pressure; fasting glucose and HgbA1c; and consider temporal artery biopsy and MRI brain and orbits.

CHAPTER RECAP

1. Four minutes of history, note started by minute eleven, education before typing.
2. Six history categories, asked out loud, because the history is scored on verbalisation.
3. Read the data actively: acuity and pupils, IOP, the relevant section, then hunt for every abnormal line.
4. Ask follow-up questions after reading the data.
5. Type specific tests with specific indications, and submit only once, because submission is final.
6. An unordered result appearing is not a sign you were wrong.
7. Education has five parts and ends with a check of understanding.
8. Name the diagnosis, then abandon the jargon entirely.
9. Dismiss the SP explicitly and only when you are truly finished with them.
10. Speak to the patient, not the screen.

CHAPTER 3

The encounter note: assessment and plan

Clinical Assessment and Interpretation 29%

Management and Documentation 25%

54% of the blueprint, captured here

ANSWER THESE COLD BEFORE READING

1. State how the assessment is entered and what the location options are.
2. State which diagnosis you enter when the patient has several findings.
3. List the five components the plan must contain.
4. State the rules for documenting a referral and a return-to-clinic date.
5. State how pediatric medication doses are documented.
6. Write a complete plan for a case in under four minutes.

WEIGHTING

54%

More than half your score is earned in this text box. Practise typing, not just thinking.

The assessment

RULES FOR THE ASSESSMENT FIELD

Rule	Detail
Selected, not typed freely	Type into the box and a drop-down populates from the Part III PEPS Diagnosis List ; select your intended diagnosis from it. Familiarise yourself with the published diagnosis list before exam day , so you know the wording the platform uses.
Location is mandatory	Options are OD, OS, OU, and N/A . Location refers to where the condition manifests in the eye or eyes. Use N/A when the diagnosis is not an ocular structure diagnosis , for example a systemic condition.
Choose the most significant diagnosis of today's visit	Informed by the chief complaint, the HPI, the history, and the clinical data. Not the incidental finding, and not a list of everything you noticed.
Be as specific as the data allows	"Primary open-angle glaucoma, OU" rather than "glaucoma". "Rhegmatogenous retinal detachment, OD" rather than "retinal detachment". Specificity is what the assessment domain rewards.

ASSESSMENT ERRORS

- **Choosing a symptom instead of a diagnosis.** "Blurred vision" or "floaters" is not an assessment.
- **Choosing the incidental finding.** A patient with a chief complaint of a curtain and a detachment does not have an assessment of "posterior vitreous detachment".
- **Leaving the location blank or defaulting to OU** when the disease is unilateral.
- **Being vaguer than the data supports.** If the case gives you an ischaemic CRVO, say ischaemic.
- **Hedging with two diagnoses** when the case points clearly to one. Commit.

The plan

FIVE COMPONENTS, EVERY TIME

- 1 **Treatment: medical therapy.** Drug, concentration, route, frequency, and duration, with instructions on use. Generic or brand names are both acceptable. **For a patient 18 years or younger you may write "prescribed at pediatric dose"** rather than calculating it.
- 2 **Treatment: non-medical therapy.** Refractive prescriptions written out in full, or referenced explicitly to the data ("prescribe the manifest refraction as documented", "prescribe the cycloplegic refraction OU"), with instructions on wear. Also: lid hygiene, compresses, contact lens changes, prism, occlusion, vision therapy, lifestyle change, and protective eyewear.
- 3 **Referrals. To whom, for what, and in what timeframe.** "Refer to ophthalmology" is explicitly unacceptable. "Refer to a retinal specialist for repair of a retinal detachment within 2 days" is the standard the guide sets.
- 4 **Additional testing.** Any further investigation required by the standard of care, with the rule-in or rule-out indication.
- 5 **Patient education, written in patient-centred language,** including the diagnosis and its cause or mechanism, plus the pertinent additional information you gave: home care, activity modification, alternatives, medication instructions and side effects, prognosis, warning signs, and any urgent action.
- 6 **Return to clinic.** A single specific interval: 1 day , 1 week , 1 month , 6 months . **No ranges ("3 to 6 months") and no calendar dates ("January 1").**

Punctuation, grammar, and spelling do not affect scoring, provided the documentation would be acceptable in a medical record. Content and completeness are what is scored. The note **autosaves and submits automatically** at the end of the cycle, and cannot be edited afterwards.

A reusable plan template

TYPE THIS SKELETON, THEN FILL IT

TREATMENT

Medical: [drug] [concentration] [route], [frequency], [duration]. Instructions: [how to use].

Non-medical: [spectacles / contact lens / prism / occlusion / therapy / lid hygiene / protective eyewear], with instructions.

REFERRAL

Refer to [specific specialist] for [specific purpose] within [specific timeframe].

[Method: telephoned today / letter sent.]

ADDITIONAL TESTING

[Test], indication: [rule in or rule out what].

PATIENT EDUCATION

Explained that the patient has [diagnosis], which means [plain-language mechanism].

Cause: [plain-language aetiology].

Discussed treatment: [what, how, and expected effect], including possible side effects of [x].

Home care and activity: [instructions].

Warning signs requiring immediate return: [list].

Prognosis discussed: [statement].

Alternatives discussed: [option].

Patient verbalised understanding.

RTC

[Single interval].

Six worked notes

WORKED NOTE 1 · PRIMARY OPEN-ANGLE GLAUCOMA

Assessment: Primary open-angle glaucoma · OU

TREATMENT

Medical: latanoprost 0.005% ophthalmic solution, 1 drop each eye at bedtime, continuing indefinitely. Instructions: instil one drop in each eye at night, close the lids gently and press at the inner corner for 2 minutes, wait 5 minutes between any other drops, and do not skip doses.

Non-medical: discussed selective laser trabeculoplasty as an alternative first-line option.

REFERRAL

None required at present. Will refer to a glaucoma specialist if target pressure is not achieved on maximal tolerated therapy or if progression occurs.

ADDITIONAL TESTING

Threshold visual field 24-2 both eyes, indication: establish a second reliable baseline. OCT retinal nerve fibre layer and ganglion cell analysis, indication: structural baseline. Central corneal thickness pachymetry, indication: interpret applanation pressure and assess risk. Optic disc photography, indication: baseline documentation. Gonioscopy, indication: confirm an open angle and exclude secondary mechanisms.

PATIENT EDUCATION

Explained that the patient has open-angle glaucoma, which means the drainage system inside the eye is not working efficiently, so the pressure is higher than the optic nerve can tolerate and the nerve is being slowly damaged. Cause: not fully known, but risk rises with age, family history, and higher pressure. Explained that the damage is painless and affects side vision first, which is why it is usually found on examination rather than noticed. Target pressure discussed. Treatment explained as above, including that the drop may cause redness, longer and darker lashes, and gradual darkening of the iris colour. Emphasised that treatment prevents further loss but cannot restore vision already lost, and that daily adherence is what protects the remaining vision. Warning signs requiring immediate return: pain, marked redness, or sudden change in vision. Advised that first-degree relatives should have their pressures and optic nerves checked. Patient verbalised understanding and restated the dosing.

RTC

6 weeks.

WORKED NOTE 2 · RHEGMATOGENOUS RETINAL DETACHMENT**Assessment:** Rhegmatogenous retinal detachment · OD**TREATMENT**

Medical: none. Nil by mouth in anticipation of surgery.

Non-medical: instructed the patient to limit activity and to position so that the superotemporal break is dependent, to limit progression of fluid toward the central vision.

REFERRAL

Refer to a vitreoretinal surgeon for surgical repair of a macula-on rhegmatogenous retinal detachment with a superotemporal horseshoe tear, within 24 hours. Telephoned the on-call retina service today and spoke with the surgeon; patient is being seen this afternoon. Written summary and images sent with the patient.

ADDITIONAL TESTING

Scleral indented dilated fundus examination of the fellow eye, indication: identify any additional break requiring prophylactic treatment. B-scan ultrasonography if the view is obscured, indication: define the extent of detachment. Fundus photography, indication: documentation.

PATIENT EDUCATION

Explained that the retina, the light-sensitive lining at the back of the eye, has a tear and is peeling away, and that fluid is collecting underneath it. Cause: the jelly inside the eye has pulled away with age and short-sightedness, and in doing so it tore the retina. Explained that this requires surgery urgently, today, because the detachment has not yet reached the central vision and repairing it before it does gives the best chance of good vision. Explained the positioning instruction and the reason for it, and that he should not eat or drink until the surgical team has assessed him. Explained that the fellow eye is at increased risk and will be examined and monitored. Warning signs: any further loss of vision before he is seen, in which case he should go directly to the emergency department. Patient verbalised understanding and repeated the positioning instruction.

RTC

1 day.

WORKED NOTE 3 · BACTERIAL CORNEAL ULCER, CONTACT LENS RELATED**Assessment:** Bacterial keratitis · OS**TREATMENT**

Medical: moxifloxacin 0.5% ophthalmic solution, 1 drop OS every hour around the clock for 48 hours, then taper according to response. Homatropine 5% ophthalmic solution, 1 drop OS three times daily for pain and to prevent posterior synechiae. Oral acetaminophen as needed for pain. No topical corticosteroid at this stage.

Non-medical: **discontinue all contact lens wear immediately**; discard the current lenses, case, and solution; do not resume wear until authorised.

REFERRAL

Refer to a corneal specialist within 24 hours if the infiltrate enlarges, if there is corneal thinning, or if there is no improvement at the 24-hour review.

ADDITIONAL TESTING

Corneal scraping from the edge and base of the ulcer for Gram stain and culture on blood, chocolate, and Sabouraud agar plus non-nutrient agar with an E. coli overlay, indication: identify the organism including Acanthamoeba and fungi. Culture of the contact lens, case, and solution, indication: identify the source organism. Corneal sensation testing (CN V assessment), indication: exclude a herpetic or neurotrophic component. Anterior segment photography, indication: document size and depth for

comparison.

PATIENT EDUCATION

Explained that there is an infected ulcer on the clear front window of the eye, caused by bacteria that got in under the contact lens, and that sleeping in lenses and rinsing them in tap water greatly increase this risk. Explained that this is serious and can scar the cornea and permanently reduce vision if not treated intensively, which is why the drops are needed every hour, including through the night, and why I need to see her again tomorrow. Explained the cycloplegic drop will blur her near vision and enlarge the pupil for a day or so. Discussed lens hygiene in detail for the future: no water contact of any kind, rub and rinse, replace the case every three months, no overnight wear, and never top off solution. Warning signs requiring immediate return: worsening pain, increasing redness, more discharge, or any further drop in vision. Advised she must not drive while the vision is reduced. Patient verbalised understanding of the hourly dosing including overnight.

RTC

1 day.

WORKED NOTE 4 · REFRACTIVE CASE WITH ACCOMMODATIVE ESOTROPIA IN A CHILD

Assessment: Refractive accommodative esotropia · OU

TREATMENT

Medical: none.

Non-medical: prescribe the **full cycloplegic refraction as documented in the clinical data, OU**, in polycarbonate lenses, for full-time wear. Do not reduce the plus.

Spectacles to be worn from waking to bedtime including for all near work.

REFERRAL

Refer to a paediatric ophthalmologist for co-management within 4 weeks if the deviation is not fully eliminated by the full hyperopic correction, for consideration of surgery for any residual angle.

ADDITIONAL TESTING

Cover test at distance and near with and without the correction, indication: quantify the residual deviation on correction. Stereoacuity testing, indication: assess binocular function. Monocular acuity with an age-appropriate optotype, indication: detect amblyopia. Dilated fundus examination, indication: exclude a sensory cause of the deviation such as an optic nerve or macular abnormality. AC/A determination, indication: identify a high AC/A that would require a near add.

PATIENT EDUCATION

Explained to the parent that the child is significantly long-sighted, and that the amount of focusing effort needed to see clearly is pulling the eye inward. Cause: the eye is shorter than average, so it has to focus hard even for distance, and focusing and turning inward are linked in the brain. Explained that glasses do the focusing work so the eye no longer needs to turn in, and that the eye turn should improve substantially or resolve with full-time wear. Emphasised that the glasses must be worn all day, that the eye will turn in again when they are removed and this is expected, and that under-correcting the glasses would allow the turn to return. Explained the risk of amblyopia in the turning eye and that we will monitor acuity, and that patching may be needed. Explained that the prescription will be rechecked regularly as the child grows. Warning signs: worsening or constant turn, or the child covering or closing one eye. Parent verbalised understanding of the requirement for full-time wear.

RTC

6 weeks.

WORKED NOTE 5 · NEURO-OPHTHALMIC CASE REQUIRING SAME-DAY ACTION

Assessment: Giant cell arteritis · N/A (with arteritic anterior ischaemic optic neuropathy, OS)

TREATMENT

Medical: **oral prednisone 60 mg (1 mg/kg) given today as the first dose**, with immediate arrangement for intravenous methylprednisolone 1 g daily for 3 days given the visual loss. Gastric protection with a proton pump inhibitor. Treatment started before biopsy.

Non-medical: advised the patient not to drive.

REFERRAL

Refer to the emergency department today, immediately, for admission, intravenous methylprednisolone, and urgent temporal artery biopsy. Telephoned the on-call medical team and the rheumatology service today and spoke directly with the physician. Also refer to rheumatology for ongoing steroid management and consideration of tocilizumab as a steroid-sparing agent.

ADDITIONAL TESTING

Westergren ESR, indication: rule in giant cell arteritis. CRP, indication: same. Platelet count and CBC, indication: thrombocytosis and anaemia support the diagnosis. Temporal artery biopsy, indication: confirm the diagnosis histologically; remains informative for approximately 2 weeks after starting steroid. Threshold visual field both eyes, indication: baseline. OCT retinal nerve fibre layer, indication: baseline. Blood pressure, fasting glucose, and HgbA1c, indication: vascular risk and steroid monitoring.

PATIENT EDUCATION

Explained that the loss of vision has been caused by inflammation of the blood vessels that supply the optic nerve, a condition called giant cell arteritis, and that it is a whole-body condition, not just an eye condition. Cause: the immune system is attacking the walls of medium-sized arteries. Explained that this is an emergency because the **other eye is at high risk of losing vision within days** if it is not treated immediately, and that high-dose steroid started today is what protects it. Explained that the vision already lost in the left eye is unlikely to recover. Explained the steroid side effects to expect and that she must not stop it herself. Explained the biopsy and that it can still be done after starting treatment. Warning signs requiring immediate return or emergency attendance: any change in the vision of the right eye, double vision, or severe headache. Advised not to drive. Patient verbalised understanding of the urgency and is being taken to the emergency department now.

RTC

1 day.

WORKED NOTE 6 · SYSTEMIC DISEASE CASE

Assessment: Severe non-proliferative diabetic retinopathy with centre-involving diabetic macular oedema · OU

TREATMENT

Medical: intravitreal anti-VEGF therapy (aflibercept) to be initiated for centre-involving macular oedema, by the retina service. No topical therapy indicated.

Non-medical: **do not prescribe new spectacles at this time** because the refraction is unstable at an HgbA1c of 9.4%; recheck when glycaemic control is stable.

REFERRAL

Refer to a retinal specialist for intravitreal anti-VEGF therapy for centre-involving diabetic macular oedema, within 2 weeks. Refer to the primary care physician within 1 week for intensification of glycaemic and blood pressure management; written summary sent describing an HgbA1c of 9.4%, blood pressure of 148/88, and severe non-

proliferative retinopathy in both eyes, and requesting reassessment of therapy.

ADDITIONAL TESTING

Macular OCT both eyes, indication: quantify central subfield thickness and monitor response. Fundus photography, indication: document severity for comparison. Fluorescein angiography or OCT angiography, indication: assess macular ischaemia and identify areas of capillary non-perfusion. HgbA1c and lipid panel, indication: systemic risk assessment. Urine albumin-to-creatinine ratio, indication: screen for nephropathy. Blood pressure, indication: independent risk factor for progression.

PATIENT EDUCATION

Explained that diabetes has damaged the small blood vessels at the back of both eyes, that these vessels are leaking fluid into the central part of the retina, which is why reading has become harder, and that the changes are now at the stage where about half of people go on to grow abnormal new blood vessels within a year if nothing changes. Cause: prolonged high blood sugar damages the walls of the smallest vessels. Explained that injections into the eye are the most effective treatment for the swelling, and what the injection involves. Explained clearly that **improving blood sugar and blood pressure control changes the course of this disease** and is as important as the injections, and that I have written to his physician today. Explained why I am not changing his glasses yet. Warning signs requiring urgent return: a sudden shower of floaters, a curtain across the vision, or a sudden drop in vision, which could mean bleeding or a detachment. Discussed that he should tell me if he becomes pregnant is not applicable; advised annual dilated examination is now insufficient and he needs review every 3 to 4 months. Patient verbalised understanding.

RTC

1 month.

PLAN ERRORS THAT COST THE MANAGEMENT AND DOCUMENTATION DOMAIN

- **A drug with no concentration, frequency, or duration.**
- **"Refer to ophthalmology"** with no subspecialty, reason, or timeframe. This is named as unacceptable in the candidate guide.
- **A range for the return-to-clinic interval**, or a calendar date.
- **No patient education section**, or a section that simply says "educated patient".
- **No warning signs** for a condition that can deteriorate.
- **Missing the systemic referral** in a case where the eye finding reveals systemic disease.
- **Failing to document that treatment was started** when you started it, and the time.
- **Writing "refractive correction" without either the numbers or an explicit reference** to the manifest or cycloplegic data.
- **Forgetting non-medical therapy** entirely: lid hygiene, lens discontinuation, positioning, prism, occlusion, protective eyewear, and lifestyle change all count.
- **Running out of time.** If you have two minutes left, type in this order: treatment, referral with timeframe, RTC, then education. Those four earn the most.

RETRIEVAL PRACTICE

1. How is the assessment entered, and what are the four location options?
2. When do you select N/A for location?
3. Which diagnosis do you enter when the patient has three findings?
4. List the six things a plan must contain.
5. Give the required structure of a referral, and quote an unacceptable version.
6. How do you document a medication for a 9-year-old?

7. How do you document a refractive prescription?
8. What are the two rules about the return-to-clinic entry?
9. Do spelling and grammar affect your score?
10. Can you edit the note after the cycle ends?
11. With two minutes left, in what order do you type?
12. Write the referral line for a 26-year-old with a macula-on retinal detachment.
13. Write the referral line for a 78-year-old with suspected giant cell arteritis.
14. Name five things that count as non-medical therapy.

KEY

1. Typed into a box that populates a drop-down from the **Part III PEPS Diagnosis List**, from which you select. Locations: **OD, OS, OU, N/A**.
2. When the diagnosis does not relate to the eye or an ocular structure, for example a systemic condition.
3. The **most significant diagnosis found at today's visit**, informed by the chief complaint, HPI, history, and clinical data.
4. Medical therapy; non-medical therapy; referrals; additional testing; patient education in patient-centred language; and a return-to-clinic interval.
5. To whom, for what, and by when. Unacceptable: **"Refer to ophthalmology."** Acceptable: "Refer to a retina specialist for the repair of a retinal detachment within 2 days."
6. **"Prescribed at pediatric dose"** is acceptable for a patient 18 years or younger.
7. Either write the numbers out, or reference the specific data explicitly ("prescribe the cycloplegic refraction as documented, OU"), with instructions on wear.
8. **No ranges** and **no calendar dates**: a single specific interval such as 1 day, 1 week, 1 month, or 6 months.
9. **No**, provided the documentation would be acceptable in a medical record.
10. **No**. It autosaves and submits automatically, and cannot be edited.
11. Treatment, referral with timeframe, RTC, then patient education.
12. "Refer to a vitreoretinal surgeon for surgical repair of a macula-on rhegmatogenous retinal detachment **within 24 hours**; on-call retina service telephoned today."
13. "Refer to the emergency department **today, immediately**, for admission, intravenous methylprednisolone, and urgent temporal artery biopsy; on-call medical and rheumatology teams telephoned today. First dose of oral prednisone 60 mg given here."
14. Spectacle or contact lens prescription, lid hygiene and warm compresses, contact lens discontinuation, positioning advice, prism or occlusion, vision therapy, protective eyewear, and lifestyle change such as smoking cessation.

CHAPTER RECAP

1. The note carries 54% of the blueprint. Practise typing it under time.
2. Select the assessment from the diagnosis list and always set the location, using N/A for non-ocular diagnoses.
3. Enter the most significant diagnosis of the visit, as specifically as the data allows.
4. Every plan needs medical therapy, non-medical therapy, referrals, additional testing, education, and an RTC.
5. Referrals need a named specialist, a reason, and a timeframe.
6. Doses need concentration, frequency, and duration; pediatric doses may be written as "prescribed at pediatric dose".
7. RTC is a single interval, never a range and never a date.
8. Patient education in the note must be in patient-centred language and must include the mechanism.
9. Include warning signs for anything that can deteriorate.
10. With time running out: treatment, referral, RTC, education.

The nine clinical presentation categories

Anterior 17 • Posterior 16 • Glaucoma 14

Systemic 11 • Refraction 11 • Neuro 9

Contact lens 8 • Binocular 8 • Pediatrics 6

Every one of your ten patient encounters is drawn from these nine categories, and each version of the exam samples them according to the blueprint weights. For each category below: the cases you are most likely to meet, the history questions that separate them, the ancillary tests to order, and the plan elements that must appear. Conditions were selected for **frequency and criticality**, exactly as the blueprint describes, so expect either a typical common presentation or a pathognomonic dangerous one.

1. Anterior segment disease • weight 17

LIKELY CASES AND THEIR HANDLING

Case	History that clinches it	Plan essentials
Bacterial keratitis	Contact lens wear, overnight wear, water exposure, pain, discharge, duration in days	Culture the edge and base plus lens, case, and solution; moxifloxacin 0.5% hourly around the clock; homatropine 5%; stop lenses; no steroid; RTC 1 day ; corneal specialist within 24 hours if worsening.
Herpes simplex epithelial keratitis	Recurrent unilateral episodes, prior cold sores, reduced sensation, no discharge	Ganciclovir 0.15% gel five times daily or oral aciclovir 400 mg five times daily for 7 to 10 days; no steroid ; corneal sensation testing; RTC 3 days ; discuss long-term prophylaxis.
Herpes zoster ophthalmicus	Dermatomal rash, nasal tip vesicles, severe pain, age, immune status	Valaciclovir 1 g three times daily for 7 to 10 days (within 72 hours); topical steroid and cycloplegia if keratitis or uveitis; IOP check; neuralgia referral; recommend recombinant zoster vaccine; RTC 3 days .
Acute anterior uveitis	Recurrence, back stiffness, bowel or skin disease, sexual history, trauma	Prednisolone acetate 1% hourly plus homatropine 5% three times daily; dilated fundus examination; IOP; HLA-B27 and RPR with treponemal test if recurrent; rheumatology referral; RTC 3 days .
Scleritis	Boring night-waking pain, joint or airway or renal symptoms, prior autoimmune diagnosis	Full-dose oral NSAID escalating to prednisone; ANCA, RF and anti-CCP, ANA, ESR and CRP, urinalysis, chest imaging; urgent rheumatology referral ; RTC 3 days .
Dry eye with meibomian gland dysfunction	Burning worse on screens, fluctuating vision, medications, autoimmune symptoms including dry mouth	Lid hygiene and warm compresses; preservative-free lubricants; cyclosporine 0.05% or lifitegrast 5%; doxycycline 50 to 100 mg daily; Sjögren serology if indicated; RTC 6 weeks .

Case	History that clinches it	Plan essentials
Chemical burn	Agent, time of exposure, whether irrigated, eye protection use	Irrigate to pH 7.0 to 7.4 first ; antibiotic, cycloplegia, steroid, doxycycline and vitamin C; IOP; symblepharon lysis; RTC 1 day ; document that protection was not worn.
Recurrent corneal erosion	Pain on waking, prior abrasion, family history of map-dot-fingerprint changes	Hypertonic saline 5% nightly for months; lubricants; doxycycline plus a short steroid pulse; bandage lens; consider micropuncture or PTK; RTC 2 weeks .
Hyperacute purulent conjunctivitis	Sexual history, rapidity, volume of discharge, partner symptoms	Ceftriaxone 1 g intramuscularly plus chlamydial cover; lavage; culture; partner treatment, STI screening, and reporting ; RTC 1 day .

2. Posterior segment disease • weight 16

Case	History that clinches it	Plan essentials
Rhegmatogenous retinal detachment	Curtain or shadow, shower of floaters, flashes, myopia, prior surgery or trauma	Vitreoretinal referral within 24 hours if macula-on ; positioning with the break dependent; nil by mouth; fellow eye indented examination; RTC 1 day .
Acute PVD with a break	Sudden floaters and flashes, duration, any field loss	Laser retinopexy urgently; indented examination of both eyes; explicit symptom education; RTC 1 to 2 weeks then 4 to 6 weeks.
Neovascular AMD	Distortion, central grey patch, duration in days, smoking, family history, supplement use	Macular OCT and fluorescein angiography ; refer for intravitreal anti-VEGF within days; AREDS2 for the fellow eye; daily monocular Amsler; smoking cessation; RTC 1 month .
Central serous chorioretinopathy	Micropsia and distortion, steroid use by <i>any</i> route, stress, snoring	Stop or substitute steroid with the prescriber; OCT with choroidal thickness; observe; half-dose photodynamic therapy if persistent at 4 months; screen for sleep apnoea; RTC 6 weeks .
Branch or central retinal vein occlusion	Sudden or subacute blur, hypertension, glaucoma, oral contraceptive use, age	OCT; blood pressure, glucose, lipids; anti-VEGF referral for macular oedema; monthly undilated gonioscopy for 6 months if ischaemic ; thrombophilia screen if under 50; RTC 1 month .
Central retinal artery occlusion	Sudden painless total loss, prior amaurosis, jaw claudication, cardiac history	Emergency department or stroke centre today ; ESR, CRP, platelets over 50; carotid duplex, ECG, echocardiography; RTC 1 day .
Macular hole or epiretinal membrane	Central distortion, duration in months, prior PVD or surgery	Macular OCT; Watzke-Allen; vitreoretinal referral for surgical consideration within 4 weeks; RTC 1 month .

Case	History that clinches it	Plan essentials
Toxoplasmic retinochoroiditis	Floaters and blur, cat exposure, undercooked meat, immune status, prior episode	Toxoplasma IgG and IgM; triple therapy with folinic acid , or trimethoprim-sulfamethoxazole; oral steroid 24 to 48 hours after the antimicrobial ; RTC 1 week .
Retinitis pigmentosa	Night blindness, bumping into things, family history, hearing loss	ERG, fields, OCT, genetic testing and counselling; acetazolamide for macular oedema; low vision referral; avoid vitamin A in <i>ABCA4</i> disease; RTC 1 year .

3. Glaucoma • weight 14

Case	History that clinches it	Plan essentials
Primary open-angle glaucoma, new	Family history, race, steroid use, sleep apnoea, migraine, prior trauma, nocturnal hypotension	Pachymetry, gonioscopy, 24-2 fields, OCT RNFL and ganglion cell, disc photography; latanoprost nightly or discuss SLT as first-line (LiGHT); target IOP stated; family screening advice; RTC 6 weeks .
Acute angle closure	Severe pain, nausea and vomiting, haloes, hyperopia, anticholinergic use, dim light onset	Full protocol in order, pilocarpine only once IOP is under 40 to 50 mmHg ; bilateral laser iridotomy ; consider lens extraction (EAGLE); RTC 1 day .
Normal-tension glaucoma	Cold hands, migraine, low blood pressure or nocturnal dipping, anaemia, sleep apnoea	30% IOP reduction target (CNTGS); MRI if the picture is atypical ; 24-hour blood pressure and sleep apnoea assessment; RTC 6 weeks .
Pseudoexfoliation glaucoma	Asymmetry, poor dilation, prior cataract surgery, systemic vascular history	Gonioscopy documenting the Sampaolesi line; SLT works well ; lower target and closer follow-up; warn about surgical risks and late IOL subluxation; RTC 6 weeks .
Steroid-induced ocular hypertension	Steroid by any route including inhaled, nasal, dermatologic, or intravitreal implant	Reduce or substitute the steroid with the prescriber; loteprednol if a topical steroid is still needed; aqueous suppressants; RTC 2 weeks .
Angle recession glaucoma	Remote blunt trauma, often decades earlier, unilateral	Gonioscopy documenting recession; medical therapy (trabeculoplasty responds poorly); annual lifelong surveillance ; examine the fellow eye; RTC 6 weeks then annually.

Case	History that clinches it	Plan essentials
Neovascular glaucoma	Diabetes, prior vein occlusion, carotid disease, pain, rapid onset	Anti-VEGF plus panretinal photocoagulation referral urgently; aqueous suppressants and steroid; avoid miotics and prostaglandins ; RTC 1 day .
Progressing glaucoma at "acceptable" pressure	Adherence, instillation technique, cost, missed doses, diurnal variation	Assess adherence and technique before escalating ; lower the target; add or switch therapy or offer SLT; consider diurnal curve; RTC 6 weeks .

4. Systemic disease • weight 11

Case	History that clinches it	Plan essentials
Diabetic retinopathy with macular oedema	Duration of diabetes, most recent HgbA1c, blood pressure, renal status, pregnancy	OCT, photography, angiography if needed; refer for anti-VEGF within 2 weeks ; write to the physician about control ; do not prescribe glasses if control is unstable; RTC 1 month .
Hypertensive retinopathy	Blood pressure control, adherence, headache, chest pain, breathlessness, pregnancy	Measure blood pressure in the room ; grade 4 with disc oedema means the emergency department today ; otherwise urgent physician referral; RTC 1 month .
Thyroid eye disease	Smoking, thyroid status and treatment, radioactive iodine, diplopia, pain, colour change	Clinical activity score; smoking cessation ; lubrication and head elevation; selenium in mild disease; urgent orbital imaging and decompression if compressive optic neuropathy; endocrinology referral; RTC 1 month .
Giant cell arteritis	Age, jaw claudication, scalp tenderness, new headache, polymyalgia, weight loss, prior amaurosis	ESR, CRP, platelets; start high-dose steroid today, before biopsy ; emergency department and rheumatology; RTC 1 day .
Sarcoidosis with uveitis	Cough, breathlessness, skin lesions, joint pain, ethnicity, prior chest imaging	ACE and lysozyme, chest imaging, calcium, interferon-gamma release assay, RPR with treponemal test; topical and systemic steroid; rheumatology or pulmonology referral; RTC 1 week .
HIV with CMV retinitis	HIV status, CD4 count, antiretroviral adherence, floaters, painless blur	Valganciclovir plus intravitreal therapy for zone 1; same-day infectious disease referral ; watch for immune recovery uveitis; RTC 1 week .
Undiagnosed diabetes or hypertension found on examination	Polyuria, thirst, weight change, family history, last physical examination	Blood pressure today ; refer for fasting glucose, HgbA1c, and lipids within 1 week with a written description of the retinal findings ; RTC 1 month .

Case	History that clinches it	Plan essentials
Obstructive sleep apnoea with floppy eyelid	Snoring, witnessed apnoea, daytime somnolence, morning irritation, sleeping position	Nightly ointment and shield; refer for a sleep study; explain the cardiovascular, NAION, and glaucoma links; RTC 6 weeks.

5. Refraction · weight 11

Case	History that clinches it	Plan essentials
Uncorrected hyperopia with asthenopia	Headache with near work, blur that clears on looking away, worse late in the day	Cycloplegic refraction; prescribe, written out or referenced to the data, with wear instructions; explain the mechanism; RTC 6 weeks.
Presbyopia	Working distance, occupation, monitor distance, hobbies, prior add, current spectacles	Determine the add by at least two methods; verify the range of clear vision brackets the actual task; choose the design for the task; explain adaptation; RTC 2 to 4 weeks if a new progressive design.
Anisometropia with aniseikonia	Pulling sensation, difficulty judging steps, prior surgery, when it started	Reduce the disparity: adjust base curve, thickness and vertex distance, or fit contact lenses; consider slab-off for vertical imbalance at near; RTC 4 weeks.
Progressing myopia in a child	Rate of change, parental myopia, outdoor time, near work habits, age of onset	Full correction, never undercorrected; axial length if available; offer orthokeratology, dual-focus soft lenses, defocus spectacles, or low-dose atropine; at least 2 hours outdoors daily; explain the long-term risk rationale; RTC 6 months.
Refractive instability in diabetes	HgbA1c, recent change in medication or control, fluctuation through the day	Do not prescribe; explain why; refer for control optimisation; RTC 4 to 6 weeks once control is stable.
Spectacle non-adaptation	Exactly what is wrong and when; what changed from the previous pair	Verify against prescription and ANSI tolerances; measure monocular PDs and heights; check base curve, material, tilt, and vertex; re-refract; check binocularity and the ocular surface; RTC 2 weeks.
Post-refractive-surgery patient	Type and date of surgery, pre-surgical refraction, night vision, dryness, fluctuation	Topography; dry eye management; document and supply pre-surgical data for future IOL calculation; explain the IOP measurement caveat; RTC 6 months.

6. Neuro-ophthalmic disease · weight 9

Case	History that clinches it	Plan essentials
Optic neuritis	Pain on eye movement, age and sex, prior neurologic episodes, heat sensitivity	MRI brain and orbits with gadolinium and fat suppression; consider intravenous methylprednisolone; never oral prednisone alone ; aquaporin-4 and MOG if atypical; neurology referral; RTC 2 weeks.
Non-arteritic AION	Noticed on waking, hypertension, diabetes, sleep apnoea, PDE5 inhibitor use, nocturnal antihypertensives	Exclude giant cell arteritis with ESR, CRP, platelets; vascular risk management; sleep study ; fellow eye education; RTC 4 to 6 weeks.
Papilloedema	Headache worse on waking and with straining, transient obscurations, pulsatile tinnitus, weight change, tetracycline or vitamin A use	Blood pressure, then same-day MRI with venography, then lumbar puncture; acetazolamide plus weight management; monitor with perimetry ; neurology or neurosurgery referral; RTC 2 weeks.
Pupil-involving third nerve palsy	Onset, pain, headache, diplopia pattern, vascular risk factors, age	Same-day CT or MR angiography for a posterior communicating artery aneurysm; telephone neurology or neurosurgery; RTC 1 day.
Horner syndrome	Pain in the neck or face , recent trauma or chiropractic manipulation, smoking, cough, weight loss; in a child, any mass	Apraclonidine to confirm, hydroxyamphetamine to localise; painful means same-day angiography of the head and neck for carotid dissection ; in a child, imaging plus urinary catecholamines for neuroblastoma; RTC 1 day.
Homonymous hemianopia	Sudden onset, other neurologic symptoms, vascular risk, bumping on one side, reading difficulty	Acute stroke pathway ; formal perimetry; driving counselling and documentation; prism and scanning rehabilitation referral; RTC 1 month.
Ocular myasthenia	Variability and fatigability, worse in the evening, swallowing or speech change, thyroid history	Ice pack and rest tests; acetylcholine receptor and MuSK antibodies; chest CT for thymoma ; thyroid function; neurology referral; warn about drugs that worsen it; do not operate ; RTC 2 weeks.

Case	History that clinches it	Plan essentials
Toxic or nutritional optic neuropathy	Ethambutol, isoniazid, linezolid, amiodarone; alcohol and tobacco; bariatric surgery; diet	Stop the offending drug with the prescriber; B ₁₂ , folate, thiamine, copper; fields and colour vision; RTC 1 month .

7. Contact lenses • weight 8

Case	History that clinches it	Plan essentials
Contact lens related infiltrative event	Wear schedule, overnight wear, replacement compliance, solution, water exposure, onset overnight	Distinguish CLARE, CLPU, infiltrative keratitis, and microbial keratitis; stop lenses; culture if there is an epithelial defect over the infiltrate; refit on daily disposables; RTC 1 day if infectious, 3 days if sterile.
Giant papillary conjunctivitis	Itching, mucus, increasing awareness and decentration late in the day, replacement interval	Evert the upper lid and grade the papillae; temporary discontinuation; change material and modality to daily disposables; mast cell stabiliser; RTC 2 weeks .
Keratoconus needing specialty fitting	Rapid prescription change, eye rubbing, atopy, family history, night driving difficulty	Topography and tomography; stop eye rubbing and treat the allergy ; scleral or hybrid lens; refer for cross-linking if progression is documented ; RTC 1 month .
Contact lens induced hypoxia	Years of wear, material, overnight wear, redness, reduced wear time	Document striae, microcysts, neovascularisation; increase Dk/t; stop overnight wear; reduce wear time; RTC 1 month .
Presbyopic contact lens fitting	Occupational distances, driving, night driving, prior monovision experience, expectations	Discuss multifocal, monovision, and modified monovision with honest trade-offs; trial period; RTC 2 weeks .
Dry eye in a contact lens wearer	End-of-day dryness, environment, medications, screen use, solution	Treat the lids and surface first; change to a daily disposable with a suitable material; preservative-free lubricants; consider scleral lenses in severe surface disease; RTC 4 weeks .
Orthokeratology or myopia control fitting	Rate of progression, parental myopia, outdoor time, hygiene reliability, parental supervision	Axial length baseline; explicit hygiene teaching including no water ; RTC 1 day after first overnight wear, then 1 week, 1 month, and 6 months.

8. Binocular vision · weight 8

Case	History that clinches it	Plan essentials
Convergence insufficiency	Asthenopia and diplopia with reading, loss of place, sleepiness, recent concussion	NPC, AC/A, vergence ranges, accommodative amplitude and facility and lag; office-based vergence and accommodative therapy with home reinforcement (CITT) ; RTC 4 weeks , reassess at 12 weeks.
Convergence excess	Near blur and headache, relief on looking up, uncorrected hyperopia	Gradient AC/A; plus add at near sized to the AC/A; correct hyperopia first; negative fusional vergence therapy; RTC 6 weeks .
Divergence insufficiency in an older adult	Distance diplopia, onset, other neurologic symptoms, headache	Exclude sixth nerve palsy, raised intracranial pressure, myasthenia, and thyroid disease ; then base-out prism at distance; RTC 4 weeks .
Intermittent exotropia	When the eye drifts, frequency, photophobia, closing one eye, family photographs	Grade control; full refractive correction; therapy, overminus, prism, or surgical referral for deteriorating control; RTC 4 months .
Superior oblique palsy	Vertical or torsional diplopia, head tilt, trauma, onset, vascular risk	Three-step test and double Maddox rod; exclude skew and myasthenia; prism; observe for recovery over months before surgery; RTC 6 weeks .
Accommodative insufficiency after concussion	Head injury timing, symptoms of photophobia, dizziness, reading intolerance, return to school or sport	Amplitude, facility, lag; plus add for near; accommodative and vergence therapy; coordinate with the concussion team on academic and sporting accommodations; RTC 4 weeks .
Vertical phoria with asthenopia	Symptoms with reading, prior prism, head posture	Associated phoria (fixation disparity) measurement and prescribe it in full; RTC 4 weeks .

9. Pediatrics · weight 6

REMEMBER: THE SP MAY BE THE PARENT

If the case is a child, the standardised patient will **tell you they are the parent or guardian**. Direct the history and, critically, the **patient education** to them as the parent of the child described. Explain in terms a parent needs: what it is, what it means for the child's vision and development, exactly what they have to do at home, how compliance will be checked, and what happens if it is not treated.

Case	History that clinches it	Plan essentials
Leukocoria	White reflex in photographs, which eye, duration, strabismus, family history, prematurity, puppy exposure	Same-week paediatric ocular oncology referral to exclude retinoblastoma; B-scan for calcification; do not treat amblyopia first; RTC 1 day to confirm the appointment.
Accommodative esotropia	Age of onset, whether intermittent or constant, worse when tired or at near, family history	Full cycloplegic plus, never undercorrected , polycarbonate, full-time wear; bifocal add if a high AC/A; amblyopia assessment; dilated fundus to exclude a sensory cause; RTC 6 weeks .
Anisometropic amblyopia	Failed school screening, no symptoms, birth history, family history	Cycloplegic refraction and full correction; refractive adaptation for up to 30 weeks before patching ; then 2 hours daily for moderate or 6 for severe, or atropine; taper and follow for a year; RTC 6 to 12 weeks .
Congenital nasolacrimal duct obstruction	Tearing from birth, discharge, absence of photophobia , corneal clarity	Crigler massage and hygiene; topical antibiotic only if purulent; probe at 12 months if unresolved; distinguish from congenital glaucoma; RTC 2 months .
Congenital glaucoma	Epiphora, photophobia, and blepharospasm , corneal enlargement or clouding	Urgent paediatric glaucoma referral for goniotomy or trabeculotomy ; brimonidine is contraindicated in infants; RTC 1 day .
Retinopathy of prematurity survivor	Gestational age, birth weight, oxygen therapy, prior treatment, current vision behaviour	Cycloplegic refraction, alignment, dilated fundus; correct significant error; amblyopia therapy; lifelong retinal surveillance; RTC 6 months .
Juvenile idiopathic arthritis uveitis	Arthritis subtype, ANA status, age of onset, absence of symptoms	Topical steroid at the lowest effective dose plus methotrexate, escalating to adalimumab ; paediatric rheumatology; screening every 3 months ; RTC 1 week then 3 months.
Vision screening failure with normal eyes	Reading difficulty, school performance, family history of learning difficulty	Cycloplegic refraction, binocular and accommodative testing; treat what is measurable; state plainly that vision therapy does not treat dyslexia ; refer for educational assessment; RTC 6 weeks .

RETRIEVAL PRACTICE: NAME THE PLAN ESSENTIALS

1. A contact lens wearer with a central infiltrate and an epithelial defect.
2. A 74-year-old with a chalky white swollen disc and jaw claudication.
3. A 3-year-old with **+5.00 D** hyperopia and a **30 Δ** esotropia.
4. A 22-year-old with a curtain in the superior field and a horseshoe tear.
5. A 55-year-old with an ischaemic CRVO.
6. A 45-year-old with a serous macular elevation on inhaled corticosteroid.

7. A 9-year-old with reading asthenopia, near exophoria, and a receded NPC.
8. A 62-year-old with progressing glaucoma at an IOP of 17 mmHg.
9. A 30-year-old with painful monocular vision loss and pain on eye movement.
10. A 4-year-old with a white pupillary reflex in one eye.
11. A 58-year-old with proptosis, lid retraction, and diplopia who smokes.
12. A 68-year-old with a new left homonymous hemianopia.
13. A patient found to have retinal microaneurysms and no known diabetes.
14. A 40-year-old with a painful Horner syndrome after neck manipulation.

KEY

1. Culture edge and base plus lens, case, and solution; moxifloxacin 0.5% hourly around the clock; homatropine 5%; stop lens wear; no steroid; **RTC 1 day**; corneal referral within 24 hours if worsening.
2. ESR, CRP, platelets; **start high-dose steroid today before biopsy**; emergency department for intravenous methylprednisolone and biopsy; rheumatology; **RTC 1 day**.
3. **Full cycloplegic plus in polycarbonate for full-time wear**, never undercorrected; amblyopia assessment; dilated fundus to exclude a sensory cause; **RTC 6 weeks**; paediatric ophthalmology if any residual angle.
4. **Vitreoretinal referral within 24 hours** for macula-on repair; position with the break dependent; nil by mouth; indented examination of the fellow eye; **RTC 1 day**.
5. OCT and anti-VEGF referral for oedema; systemic risk workup; **monthly undilated gonioscopy for 6 months** because neovascular glaucoma peaks at 3 to 4 months; **RTC 1 month**.
6. Stop or substitute the steroid with the prescriber; OCT with choroidal thickness; observe; half-dose photodynamic therapy if persistent at 4 months; screen for sleep apnoea; **RTC 6 weeks**.
7. **Office-based vergence and accommodative therapy with home reinforcement**; **RTC 4 weeks** with reassessment at 12 weeks.
8. **Assess adherence and instillation technique first**; then lower the target, add or switch therapy or offer SLT, consider a diurnal curve; **RTC 6 weeks**.
9. MRI brain and orbits with gadolinium and fat suppression; consider intravenous methylprednisolone; **never oral prednisone alone**; neurology referral; **RTC 2 weeks**.
10. **Same-week paediatric ocular oncology referral** to exclude retinoblastoma; B-scan for calcification; **RTC 1 day** to confirm the appointment.
11. Clinical activity score; **smoking cessation**; lubrication and head elevation; endocrinology; urgent imaging and decompression if compressive optic neuropathy; **RTC 1 month**.
12. **Acute stroke pathway today**; formal perimetry; driving counselling documented; rehabilitation referral; **RTC 1 month**.
13. Measure blood pressure in the room; refer to primary care **within 1 week** for fasting glucose, HgbA1c, and lipids with a written description of the findings; **RTC 1 month**.
14. **Same-day CT or MR angiography of the head and neck from the arch** for carotid dissection; telephone the stroke or neurology service; **RTC 1 day**.

CHAPTER RECAP

1. Nine categories, weighted; expect either a typical common case or a pathognomonic dangerous one.
2. Anterior segment is the heaviest category, and most of its cases are treated by you rather than referred.
3. Posterior segment cases usually hinge on one urgent decision: refer today, refer within days, or observe with education.
4. Glaucoma cases test target setting, adherence assessment, and the mechanism of closure.
5. Systemic cases test whether you measure blood pressure, write to the physician, and state the urgency.
6. Refraction cases test prescribing judgement, including when not to prescribe.
7. Neuro cases contain the same-day emergencies: giant cell arteritis, pupil-involving third nerve palsy, painful Horner, and papilloedema.
8. Contact lens cases test differentiation of infiltrative events and hygiene counselling.
9. Binocular cases test the CITT answer and the exclusion of neurologic mimics.
10. Pediatric cases may be delivered through a parent, and leukocoria and congenital glaucoma are the two emergencies.

Anterior segment skills station, item by item

Biomicroscopy · items 2–25

4-mirror gonioscopy · items 26–41

Goldmann applanation · items 42–58

General items 1 and 59

STATION RULES YOU MUST KNOW BEFORE YOU START

- **All three skills are performed on ONE eye.** Ask the SP which eye at the start of each skill, and ask again if you forget.
- **You may alter the order of the skills within the station,** and you may alter the order of items within a skill provided your sequence makes logical sense.
- **Item 1 is a stated intention, not an action:** you must *say* that you would properly disinfect the ocular instruments before using them. The gonioscopy lens and the tonometer probe are to be assumed already cleaned, but **you must still state that you would disinfect them.**
- **State every structure as you view it, and describe every finding as you would enter it into a medical record.** Half of the items on this form are "accurately describe" items, and they are earned only by speaking.
- **You are scored on what the camera captures through the LEFT ocular.** Ensure the image is completely seen through the left ocular.
- **Views must not be fleeting,** and "entire" means **360 degrees or the complete structure.**
- **Assume the patient has less than three dioptres of corneal cylinder** for tonometry, so no prism rotation is required.
- **Maintain hygiene throughout** (item 59). Touching your face or hair after washing breaks protocol.

Opening the station

FIRST 20 SECONDS OF THE CYCLE

"Good morning, Ms. Lee. I am candidate [OE TRACKER number]. Today I will be examining the front of your eye with this microscope, then looking at the drainage angle inside your eye with a small lens, and then measuring the pressure in your eye. Which eye am I examining today?"

"Before I begin, I would properly disinfect all ocular instruments prior to using them, including the biomicroscope headrest and chin rest, the gonioscopy lens, and the tonometer probe."

That second sentence earns item 1. Say it explicitly, at the start, and say it once. Do not assume that wiping the headrest during observation substitutes for saying it.

Skill 1 · Biomicroscopy (items 2 to 25)

ITEM-BY-ITEM SCRIPT AND TECHNIQUE

Item	What is scored	Technique and the words to say
2	Explain the purpose to the patient	<i>"I am going to use this microscope to look closely at the surface and the front structures of your eye. It uses a bright light but it does not touch your eye. Please put your chin on the chin rest and your forehead against the bar for me."</i>

Item	What is scored	Technique and the words to say
3	Properly examine the entire upper and lower lids and lashes	Low magnification ($10\times$), wide diffuse beam , moderate illumination. Scan the full horizontal extent of the upper lid from medial to lateral canthus, then the lower lid. Do not stop at the centre: "entire" means the whole lid.
4	Accurately describe the lids and lashes	<i>"The upper and lower lid skin is intact with no lesions, oedema, or erythema. The lashes are of normal number, direction, and distribution, with no collarettes, no scurf, no madarosis, and no trichiasis or distichiasis."</i>
5	Properly examine the entire upper and lower lid margins and the lower puncta	Increase magnification to $16\times$. Sweep the full length of both margins. Have the patient look up and slightly out, and pull the lower lid down gently to bring the lower punctum into view. The punctum is a named item: it must be seen and described.
6	Accurately describe the margins and the lower puncta	<i>"The upper and lower lid margins are smooth and regular with no notching, telangiectasia, or thickening. The meibomian gland orifices are patent and non-capped along both margins. The lower punctum is patent, of normal size, and correctly apposed to the globe."</i>
7	Properly examine the entire inferior palpebral conjunctiva using lower lid eversion	Ask the patient to look up. Place your thumb or finger below the lash line and evert the lower lid to expose the full tarsal and forniceal conjunctiva, medial to lateral.
8	Accurately describe the inferior palpebral conjunctiva	<i>"The inferior palpebral conjunctiva is pink and smooth, with no follicles, no papillae, no concretions, no membranes, and no scarring or symblepharon. The inferior fornix is of normal depth."</i>
9	Properly examine the entire superior palpebral conjunctiva using upper lid eversion	This is the item candidates most often lose. Instruct clearly, then evert properly. <i>"I am going to turn your upper eyelid out for a moment to look underneath it. It will feel odd but it should not hurt. Please look all the way down and keep looking down."</i> Grasp the lashes and the lid margin, pull the lid down and away from the globe, then use a cotton-tipped applicator placed at the upper lid crease to rotate the tarsus outward. You are explicitly permitted to use a cotton-tipped applicator. Hold the eversion long enough to sweep medial to lateral. Then release gently and ask the patient to look up and blink.
10	Accurately describe the superior palpebral conjunctiva	<i>"The superior palpebral conjunctiva is pink and smooth with no follicles, no papillae greater than 0.3 millimetres, no giant papillae, no concretions, no membranes, no scarring or Arlt line, and no foreign body in the subtarsal sulcus."</i>
11	Properly examine the entire bulbar conjunctiva and sclera	Diffuse beam, moderate magnification. Direct the patient into all four gaze positions to expose the full bulbar surface: <i>"Please look up for me. Now down. Now toward your nose. Now toward your ear."</i> Include the limbus circumferentially.
12	Accurately describe the bulbar conjunctiva and sclera	<i>"The bulbar conjunctiva is clear and quiet, with no injection, no chemosis, no follicles, no haemorrhage, and no pinguecula or pterygium. The limbus is intact with clear palisades of Vogt. The underlying sclera is white with no thinning, no nodules, no episcleral or deep vascular injection, and no pigmentation."</i>

Item	What is scored	Technique and the words to say
13	Properly examine the entire cornea using a parallelepiped	Beam width about 2 to 3 mm, angle 45 degrees, magnification 16×. Sweep the parallelepiped systematically across the entire cornea, limbus to limbus, in overlapping vertical or horizontal passes. Announce that you are doing so: <i>"I am now examining the entire cornea with a parallelepiped, sweeping from limbus to limbus."</i>
14	Properly examine the entire cornea using an optic section	Narrow the beam to the thinnest slit and increase the angle. Sweep the entire cornea again , this time announcing the layers: <i>"I am now using an optic section to assess corneal depth and to localise any findings by layer, again sweeping the entire cornea."</i> Both item 13 and item 14 require the entire cornea, so you sweep twice.
15	Accurately describe the cornea by layer: epithelium, stroma, endothelium	Describe all three layers explicitly. <i>"The epithelium is intact, smooth, and clear with no punctate staining, no defect, and no oedema. The stroma is clear with no infiltrate, no scar, no neovascularisation, no thinning, and no striae. Descemet membrane and the endothelium appear clear with no guttae, no keratic precipitates, and no pigment deposition."</i>
16	Properly assess the temporal anterior chamber angle using the Van Herick technique	Narrow parallelepiped or optic section, beam at 60 degrees from the observation axis, placed at the temporal limbus , magnification 16×. Announce it: <i>"I am now performing a Van Herick assessment at the temporal limbus, comparing the depth of the aqueous shadow with the corneal section thickness."</i>
17	Accurately describe the temporal Van Herick finding	Use the grading explicitly. <i>"The shadow depth is greater than one half of the corneal section thickness, which is a Van Herick grade 4, a wide open angle temporally."</i> The grades: 4 shadow greater than 1/2 corneal thickness; 3 shadow between 1/4 and 1/2; 2 shadow 1/4; 1 shadow less than 1/4; grades 1 and 2 are potentially occludable.
18	Properly examine the anterior chamber for cell and flare	Small, bright, conical beam (about 1 mm square or a small round aperture), maximum illumination , high magnification, beam angled at about 45 to 60 degrees, focused in the aqueous midway between the corneal endothelium and the anterior lens surface , in a darkened room . Announce it: <i>"I am darkening the room and using a small, bright conical beam focused in the aqueous between the cornea and the lens to assess for cells and flare."</i>
19	Accurately describe cell and flare	Use SUN grading. <i>"There are no cells and no flare in the anterior chamber: grade zero cells and grade zero flare, with an optically clear aqueous."</i> If present, grade it: 0.5+ is 1 to 5 cells, 1+ is 6 to 15, 2+ is 16 to 25, 3+ is 26 to 50, 4+ is over 50 cells in a 1 by 1 mm field.
20	Properly examine the entire iris	Return to a wider beam and moderate illumination. Sweep the full iris surface, 360 degrees , from the pupil margin to the root, in overlapping passes. Move the patient's gaze to see the far periphery.
21	Accurately describe the iris	<i>"The iris is flat and of normal architecture and colour, with an intact collarette and normal crypts and contraction furrows. There are no naevi, no Lisch nodules, no Koepe or Busacca nodules, no neovascularisation at the pupil margin, no transillumination defects, no atrophy, and no synechiae. The pupil is round, centred, and of normal size, with an intact pupillary ruff."</i>

Item	What is scored	Technique and the words to say
22	Properly perform transillumination of the entire iris	This is a distinct technique, not the same as item 20. Place a small, bright beam directly through the pupil, coaxially, to retro-illuminate the iris from behind using the fundus reflex, and move the beam across the pupil to transilluminate the whole iris , 360 degrees. Announce it: <i>"I am now transilluminating the entire iris by directing the beam through the pupil to retro-illuminate the iris and look for transillumination defects."</i>
23	Accurately describe iris transillumination	<i>"On transillumination the entire iris is intact with no transillumination defects: no mid-peripheral radial spoke-like defects and no moth-eaten peripupillary defects."</i>
24	Examine the ocular tissues in a smooth, logical sequence	Move front to back and outside in : lids and lashes, margins and puncta, palpebral conjunctiva (lower then upper), bulbar conjunctiva and sclera, cornea (parallelepiped then optic section), Van Herick, anterior chamber, iris, iris transillumination. Do not jump back and forth, do not re-do structures, and do not change magnification and illumination more than necessary.
25	Properly instruct the patient throughout	Give a clear instruction before every change: <i>"Look up for me. Now down. Look at my ear. I am going to turn your lid out now, please keep looking down. I am going to make the light brighter, this may feel uncomfortable but it will only be a moment. You can blink normally whenever you need to."</i> Warn before every bright light and before every touch.

Skill 2 · Four-mirror gonioscopy (items 26 to 41)

NON-NEGOTIABLES FOR THIS SKILL

- **State that you would disinfect the lens** even though you assume it is clean.
- **Anaesthetic in the examined eye only.**
- **A cushioning solution such as Celluvisc must be used** on the lens for patient comfort. This is explicitly required by the candidate guide.
- **All four quadrants**, each with a clear, centred, maintained view of the *entire* quadrant, and each correctly named.
- **Identify the most posterior visible angle structure in every quadrant**, and the **iris approach in the inferior quadrant** (item 31 asks for quadrant, most posterior structure, *and* iris approach).
- **Maintain control of the lens at all times** (item 38): never let go, never let it slide uncontrolled, keep a stable brace on the patient's cheek or the slit lamp.
- **Assess and describe corneal integrity *without staining at the conclusion*** (item 40): use the white light and a parallelepiped, not fluorescein.

ITEM-BY-ITEM SCRIPT AND TECHNIQUE

Item	What is scored	Technique and the words to say
26	Explain the purpose, including the drop	<i>"Next I am going to look at the drainage angle inside your eye, which is where fluid leaves the eye. To do that I need to place a small lens gently on the front of your eye. First I will put in an anaesthetic drop so that it is comfortable and you will not feel the lens. The drop may sting for a second or two."</i> Naming the drop is part of the item.

Item	What is scored	Technique and the words to say
27	Properly prepare the lens and instil anaesthetic into the appropriate eye	State the disinfection intention, then apply the cushioning solution to the lens face: <i>"I would disinfect the lens prior to use. I am applying a cushioning solution to the lens surface for your comfort."</i> Instil proparacaine 0.5% into the examined eye only , without touching the lids or lashes with the bottle, and confirm which eye out loud.
28	Provide proper instructions to the patient	<i>"Please put your chin on the rest and your forehead forward. I want you to keep both eyes open and look straight ahead at the light in front of you. Keep breathing normally, try not to squeeze, and tell me if you are uncomfortable at any point."</i>
29	Insert and use the lens safely and efficiently	Low magnification, short vertical beam, moderate illumination, room lights low. Approach from below or from the side, brace your hand against the patient's cheek or the headrest , and place the lens gently and squarely on the cornea without pressing. Avoid the lashes. Do not indent unless you intend to.
30	Clear, maintained image of the entire inferior quadrant	The superior mirror views the inferior angle . Centre the view, keep the beam perpendicular to the mirror, and sweep the full extent of the quadrant. Hold it: a fleeting view does not count . Ask if needed: <i>"Do you have a view?"</i>
31	Accurately identify the inferior quadrant , the most posterior visible structure , and the iris approach	<i>"I am viewing the inferior angle through the superior mirror. The most posterior structure visible is the ciliary body band. The iris approach is regular and flat with a normal insertion, and the angle is wide open, graded Shaffer 4. There are no peripheral anterior synechiae, no neovascularisation, and trabecular pigmentation is grade 1."</i>
32–33	Nasal quadrant: view and identification	Rotate the lens 90 degrees. The temporal mirror views the nasal angle . <i>"I am now viewing the nasal angle. The most posterior visible structure is the ciliary body band, the angle is open, Shaffer grade 4, with grade 1 pigmentation and no synechiae."</i>
34–35	Superior quadrant: view and identification	The inferior mirror views the superior angle . The superior angle is normally the narrowest, so expect a different answer. <i>"I am now viewing the superior angle. The most posterior visible structure is the anterior trabecular meshwork, the angle is narrower here, Shaffer grade 2 to 3, with grade 2 pigmentation and no synechiae."</i>
36–37	Temporal quadrant: view and identification	The nasal mirror views the temporal angle . <i>"I am now viewing the temporal angle. The most posterior visible structure is the ciliary body band, Shaffer grade 4, grade 1 pigmentation, no synechiae."</i>
38	Maintain control of the lens at all times	Keep three points of contact: fingers on the lens, hand braced on the cheek or headrest, and the patient's head stable against the bar. Never release the lens while it is on the eye, and never let it slide across the cornea uncontrolled.
39	Remove the lens safely and efficiently	Warn the patient, then lift the lens straight off without dragging. <i>"I am taking the lens off now, please keep looking straight ahead."</i> Do not ask the patient to blink or squeeze it off.

Item	What is scored	Technique and the words to say
40	Assess and describe corneal integrity without staining	Return to the slit lamp, white light, parallelepiped, and re-examine the cornea. <i>"Following lens removal I am reassessing corneal integrity without stain. The epithelium is intact and smooth with no abrasion, no epithelial disruption, and no oedema."</i> Do not use fluorescein for this item.
41	Smooth and logical sequence	Explain, anaesthetise, prepare, instruct, insert, then go around the angle in one continuous rotation (for example inferior, nasal, superior, temporal), remove, then reassess the cornea. Do not revisit quadrants.

MIRROR TO QUADRANT, AND THE TWO GRADING SYSTEMS

The mirror is opposite the angle you see. Superior mirror shows the inferior angle; inferior mirror shows the superior angle; nasal mirror shows the temporal angle; temporal mirror shows the nasal angle.

Angle structures, anterior to posterior: Schwalbe line, trabecular meshwork, scleral spur, ciliary body band, iris root.

Shaffer grading (angular width): grade 4 is **35 to 45 degrees** and wide open; grade 3 is **20 to 35 degrees**; grade 2 is **20 degrees** and narrow; grade 1 is **10 degrees** and extremely narrow; grade 0 is closed. **Spaeth** additionally records iris insertion, angular width, and iris configuration.

Skill 3 · Goldmann applanation tonometry (items 42 to 58)

ITEM-BY-ITEM SCRIPT AND TECHNIQUE

Item	What is scored	Technique and the words to say
42	Explain the purpose, including the drop	<i>"Now I am going to measure the pressure inside your eye. I will put in a drop that has an anaesthetic and a yellow dye in it, so the eye is numb and I can see the measurement. The dye may make your vision look a little yellow for a few minutes. A small blue-lit probe will touch the front of your eye very gently; you should not feel it."</i>
43	Instil the anaesthetic and fluorescein combination properly in one eye	State the disinfection intention for the probe first. Then instil the combination drop (benoxinate with fluorescein, or proparacaine plus a fluorescein strip) into the examined eye only , without touching the eye or lashes. Confirm the eye out loud. Ask the patient to blink a few times to distribute the dye.
44	Adjust the illumination system properly	Cobalt blue filter, widest beam, full illumination , illumination arm swung about 60 degrees temporally from the observation axis so the beam floods the prism. Magnification low (10×). Say what you are doing: <i>"I am setting the cobalt blue filter, the widest beam at full illumination, with the illumination arm at approximately 60 degrees."</i>
45	Assess and describe the cornea for staining before tonometry	Look with the blue light before you touch the eye. <i>"Before tonometry I am assessing the cornea with the cobalt blue filter. There is no punctate staining, no epithelial defect, and no abrasion; the epithelium is intact."</i> This is a separate scored item from item 56, which is the post-tonometry assessment.

Item	What is scored	Technique and the words to say
46	Assess and describe the quality of the tear film before tonometry	<i>"The tear film is of adequate volume with a normal meniscus height and no excess fluorescein pooling, no debris, and a smooth even distribution."</i> Excess fluorescein widens the mires and overestimates pressure; too little narrows them and underestimates it, so this assessment has a direct measurement purpose.
47	Set the probe pressure at an appropriate level	Set the dial to about 1 , meaning 10 mmHg , before contact. <i>"I am pre-setting the dial to 10 millimetres of mercury."</i> Starting at zero risks a sudden jump; starting very high risks corneal indentation.
48	Properly position the tonometer and align the probe	Swing the tonometer into place with the prism set at 0 degrees on the graduation (valid for under three dioptres of corneal cylinder, which you are told to assume). Align the prism centre to the corneal apex, at the same height as the patient's pupil, and centre it in the beam.
49	Provide proper instructions, including a fixation target	Naming the fixation target is part of the item. <i>"Please keep both eyes wide open and look straight ahead past my ear at the corner of the wall. Try not to squeeze or hold your breath. Keep looking at that spot and breathe normally. It will touch very gently and you should not feel it."</i>
50	Applanate safely and efficiently	Watch from the side, outside the oculars , as you advance until the prism just touches, then look through the left ocular. Advance smoothly, do not push, and do not contact the lids or lashes. Retract at once if the patient squeezes.
51	Adjust so the semicircles are symmetrical and centred	Use the joystick to centre the mires horizontally and vertically so the two semicircles are of equal size and sit in the middle of the field . Say it: <i>"I am centring the mires and equalising the two semicircles."</i>
52	Ensure the mires are of appropriate width	Mire width should be about one tenth of the diameter of the applanated area . If they are too thick, there is too much fluorescein or tears: dry the prism and re-instil less. If too thin, add fluorescein. Say what you are judging: <i>"The mire width is appropriate, approximately one tenth of the ring diameter."</i>
53	Properly align the mires	Turn the dial until the inner edges of the two semicircles just touch , forming an even figure of eight, and take the reading at the midpoint of the pulsation . Overlapping means the reading is too low; separated means too high.
54	Safely retract the tonometer	Withdraw the joystick smoothly and swing the arm away before the patient moves. <i>"I am moving the instrument away now, you can blink."</i>
55	Accurately state the obtained finding with the time	The time is part of the item. <i>"The intraocular pressure of the right eye by Goldmann applanation is 16 millimetres of mercury at 10:42 in the morning."</i> State the eye, the method, the value, and the clock time.
56	Assess and describe the cornea using the blue filter after tonometry	Re-examine with the cobalt blue filter. <i>"Following tonometry I am reassessing the cornea with the cobalt blue filter. The epithelium remains intact with no punctate staining and no abrasion from the probe."</i>

Item	What is scored	Technique and the words to say
57	Smooth and logical sequence	Explain, disinfect statement, instil, set illumination, assess cornea, assess tear film, set the dial, position and align, instruct with a fixation target, applanate, centre and equalise, check width, align mires, read, retract, state with time, reassess the cornea, then educate.
58	Educate and explain the patient's IOP	A separate scored item, and easy to forget in the rush to finish. <i>"Your eye pressure today was 16, which is in the normal range. Eye pressure matters because pressure that is too high over time can damage the nerve at the back of the eye, which is what happens in glaucoma. Yours is healthy, and we will check it again at each visit so we can see any change over time."</i>

THE TEN ITEMS MOST COMMONLY LOST IN THIS STATION

1. **Item 1:** forgetting to say you would disinfect the instruments.
2. **Item 9:** an inadequate or absent upper lid eversion. Practise it until it takes eight seconds.
3. **Items 5 and 6:** omitting the **lower punctum**.
4. **Item 14:** doing an optic section only centrally instead of sweeping the entire cornea.
5. **Items 16 and 17:** forgetting the Van Herick assessment entirely, or not stating a grade.
6. **Items 22 and 23:** treating iris transillumination as the same as examining the iris. It is a separate technique and a separate pair of items.
7. **Item 31:** giving the quadrant and the posterior structure but forgetting the **iris approach**.
8. **Item 40:** staining the cornea to check integrity, when the item specifies **without** staining.
9. **Item 55:** stating the pressure without the **time**.
10. **Item 58:** finishing the measurement and never **explaining the pressure to the patient**.

A FIFTEEN-MINUTE PACING PLAN FOR THE ANTERIOR STATION

Clock	Task
0:00–0:30	Introduce, ask which eye, state the disinfection intention, position the patient.
0:30–6:00	Biomicroscopy. Lids and lashes, margins and puncta, lower then upper palpebral conjunctiva, bulbar conjunctiva and sclera, cornea by parallelepiped then optic section, Van Herick, cells and flare, iris, iris transillumination.
6:00–10:30	Gonioscopy. Explain and anaesthetise, prepare with cushioning solution, insert, four quadrants in one rotation, remove, reassess cornea without stain.
10:30–14:30	Applanation. Explain and instil, set illumination, pre-tonometry cornea and tear film, set dial, align, instruct with fixation target, applanate, read, retract, state with time, post-tonometry cornea, educate the pressure.
14:30–15:00	Buffer. If you are ahead, repeat any item you are unsure of, announcing it. Remain in the room until the announcement.

If you are behind at 10:30, move to tonometry anyway. Tonometry contains 17 items, and finishing it is worth more than perfecting a gonioscopy quadrant.

RETRIEVAL PRACTICE

1. What exactly earns item 1, and when do you say it?
2. Which two items require the *entire* cornea, and what is the difference between them?
3. Give the Van Herick grades and which are potentially occludable.
4. Describe the beam and room conditions for assessing cells and flare.

5. How does iris transillumination differ from examining the iris?
6. What must you do to the gonioscopy lens before insertion, and what must you state?
7. Which mirror views which quadrant?
8. Which quadrant requires you to state the iris approach as well?
9. How do you assess corneal integrity after gonioscopy, and what must you not use?
10. What illumination settings does Goldmann applanation require?
11. Which two corneal and tear assessments must you make *before* applanating, and why do they matter?
12. What is the correct mire width and mire alignment endpoint?
13. What three things must accompany the pressure value when you state it?
14. What is item 58, and why is it commonly lost?
15. You are behind schedule at 10:30 with gonioscopy incomplete. What do you do and why?

KEY

1. **Stating that you would properly disinfect the ocular instruments prior to using them.** Say it in your opening, explicitly, once.
2. Items 13 and 14. Item 13 requires a **parallelepiped** swept over the entire cornea; item 14 requires an **optic section** swept over the entire cornea. You sweep twice.
3. Grade 4: shadow greater than half the corneal section thickness. Grade 3: between a quarter and a half. Grade 2: a quarter. Grade 1: less than a quarter. **Grades 1 and 2 are potentially occludable.**
4. A **small, bright conical beam** at maximum illumination, high magnification, angled about 45 to 60 degrees, focused in the aqueous midway between the corneal endothelium and the anterior lens surface, in a **darkened room**.
5. Examining the iris uses a wider beam on the iris surface; **transillumination directs the beam through the pupil** to retro-illuminate the iris using the fundus reflex, moving the beam across the pupil to cover the entire iris and reveal transillumination defects.
6. Apply a **cushioning solution such as Celluvisc**, and instil anaesthetic in the examined eye only. You must state that you would **disinfect the lens prior to use**.
7. The mirror is **opposite** the quadrant viewed: superior mirror shows the inferior angle, inferior mirror the superior angle, nasal mirror the temporal angle, and temporal mirror the nasal angle.
8. The **inferior** quadrant, item 31: quadrant, most posterior visible structure, **and iris approach**.
9. With white light and a parallelepiped at the slit lamp, describing epithelial integrity. **Do not use fluorescein**: the item specifies without staining.
10. **Cobalt blue filter, widest beam, full illumination**, illumination arm at about 60 degrees, low magnification.
11. **Corneal staining** (item 45) and **tear film quality** (item 46). Excess fluorescein widens the mires and overestimates pressure; too little narrows them and underestimates it, and pre-existing staining must be distinguished from probe-induced staining afterwards.
12. Width about **one tenth of the diameter of the applanated ring**; the endpoint is when the **inner edges of the two semicircles just touch**, read at the midpoint of the pulsation.
13. The **eye**, the **method**, and the **time**: "IOP right eye by Goldmann applanation is 16 mmHg at 10:42."
14. **Educating and explaining the patient's IOP**. It is lost because candidates treat the reading as the end of the skill and move on.
15. **Move to tonometry**. Tonometry has 17 items and gonioscopy quadrants are worth fewer; completing the remaining skill earns more than perfecting one quadrant.

CHAPTER RECAP

1. Say you would disinfect, ask which eye, and narrate everything.
2. Half the items are "accurately describe": they are earned only by speaking, in medical record language.
3. "Entire" means 360 degrees; a fleeting view scores nothing; the left ocular is what is recorded.
4. Upper lid eversion, the lower punctum, Van Herick, and iris transillumination are the four most missed biomicroscopy elements.
5. Sweep the whole cornea twice: parallelepiped, then optic section.
6. Gonioscopy needs a cushioning solution, four centred maintained quadrant views, and the iris approach inferiorly.
7. Reassess the cornea after gonioscopy *without* stain.
8. Applanation: blue filter, widest beam, 60 degrees, dial at 10, fixation target named, mires one tenth wide and inner edges touching.
9. State the pressure with the eye, the method, and the clock time.
10. Finish by explaining the pressure to the patient. That is a scored item.

CHAPTER 6

Posterior segment skills station, item by item

Binocular indirect ophthalmoscopy · items 1–21

Dilated biomicroscopy and fundus lens · items 22–43

General item 44

STATION RULES YOU MUST KNOW BEFORE YOU START

- **All items are performed on ONE eye.** Ask which eye at the start of each skill.
- **The SP sets the BIO rheostat at their calibrated setting. Do not adjust it.**
- **Optimal working distance is 18 to 25 inches**, and the camera records the image you view through either ocular.
- **A "peripheral view" is defined:** an area of retina **extending from the equator to the ora serrata** and containing **at least a partial view of a vortex vein ampulla**. That definition is the target for every one of the eight BIO quadrant items.
- **Fill the condensing lens as much as possible without the arcades in view.** If you can see the arcades, you are too posterior for a peripheral item.
- **BIO may be performed seated or reclined**, with the patient's consent.
- **Be efficient with light.** Do not shine light on the same area for an extended period. The SP may blink or ask for a break, and you should accommodate that.
- **For retroillumination of the crystalline lens only** (item 25), the light source must be on **maximum illumination** and you must **move the light source** to obtain the retroilluminated image.
- **Maintain hygiene throughout** (item 44).

Skill 4 · Binocular indirect ophthalmoscopy (items 1 to 21)

OPENING (ITEM 1)

"Good morning, Ms. Lee. I am candidate [OE TRACKER number]. Which eye am I examining today? I am going to use this headset and a hand-held lens to look at the inside of the back of your eye, particularly the outer edges of the retina, which is the light-sensitive lining. The light is bright and it will be uncomfortable, but it does not touch your eye and it will not harm it. You can blink whenever you need to, and tell me if you need a break at any point. Would you prefer to sit up or lie back for this?"

That paragraph earns item 1, obtains consent for positioning, and pre-empts the discomfort that otherwise causes the patient to squeeze and lose you views.

THE EIGHT QUADRANTS: ITEMS 2 TO 17, IN PAIRS

Items	Quadrant	Where you stand and where the patient looks	What to say
2, 3	Superior	Stand at the patient's inferior side (below), lens over the eye, patient looks up .	<i>"Please look all the way up at the ceiling. I am now viewing the superior peripheral retina. I can see a vortex vein ampulla. The superior peripheral retina from the equator to the ora serrata is flat and intact, with no holes, tears, lattice degeneration, pigmentary change, or retinal detachment."</i>

Items	Quadrant	Where you stand and where the patient looks	What to say
4, 5	Superior-nasal	Stand inferotemporally, patient looks up and toward the nose.	<i>"Please look up and toward your nose. I am viewing the superior-nasal peripheral retina, with a vortex vein ampulla in view. It is flat and intact with no breaks or degeneration."</i>
6, 7	Nasal	Stand on the temporal side, patient looks toward the nose .	<i>"Please look toward your nose. I am viewing the nasal peripheral retina. Vortex ampulla visible. Flat and intact, no breaks, no lattice, no pigment clumping."</i>
8, 9	Inferior-nasal	Stand superotemporally, patient looks down and toward the nose.	<i>"Please look down and toward your nose. I am viewing the inferior-nasal peripheral retina, vortex ampulla in view, flat and intact with no breaks or degeneration."</i>
10, 11	Inferior	Stand at the patient's superior side (above the head), patient looks down .	<i>"Please look all the way down. I am viewing the inferior peripheral retina, vortex ampulla visible, flat and intact with no breaks, no retinoschisis, and no subretinal fluid."</i>
12, 13	Inferior-temporal	Stand superonasally, patient looks down and toward the ear.	<i>"Please look down and toward your ear. I am viewing the inferior-temporal peripheral retina, vortex ampulla in view, flat and intact."</i>
14, 15	Temporal	Stand on the nasal side, patient looks toward the ear .	<i>"Please look toward your ear. I am viewing the temporal peripheral retina, vortex ampulla visible, flat and intact with no breaks and no lattice degeneration."</i>
16, 17	Superior-temporal	Stand inferonasally, patient looks up and toward the ear.	<i>"Please look up and toward your ear. I am viewing the superior-temporal peripheral retina, vortex ampulla in view, flat and intact with no horseshoe tear or operculated hole."</i>

THE POSTERIOR POLE AND THE CLOSING ITEMS

Item	What is scored	Technique and words
18	Properly examine and obtain a clear image of the posterior pole , filling the lens as much as possible	Patient looks straight ahead at your forehead or a distance target . Note that for this item the arcades <i>should</i> be in view: the exclusion of the arcades applies only to the eight peripheral items. <i>"Please look straight ahead at my forehead. I am now viewing the posterior pole."</i>
19	Provide an accurate assessment of the posterior pole	<i>"At the posterior pole the optic nerve is pink and well perfused with distinct margins, the cup-to-disc ratio appears approximately 0.3 horizontally and 0.3 vertically, the arcades are of normal calibre and course with a normal arteriole-to-venule ratio of two to three, the macula is flat and intact with a normal foveal reflex, and the background retina is flat with no haemorrhage, exudate, or pigmentary change."</i>

Item	What is scored	Technique and words
20	Perform the skill in a safe manner for the patient	Do not brace on the patient's face or lean on them. Do not sustain the beam on one area. Warn before each move. Accept blinks and breaks. Do not touch below the shoulder. Keep the light off the fovea for prolonged periods.
21	Perform in a smooth and logical sequence	Go around the periphery in one continuous rotation , either clockwise or anticlockwise, then finish centrally. Do not skip and return.

BIO TECHNIQUE POINTS THAT DETERMINE WHETHER YOU GET A VIEW AT ALL

1. **Set the headset during observation time.** Adjust the pantoscopic tilt so the mirrors align with your pupils, set the ocular separation, and confirm you can see clearly at **18 to 25 inches**. Do this before the cycle starts.
2. **Stand opposite the quadrant you want.** To see the superior retina you stand below and the patient looks up. The lens must be positioned so the light enters and returns through the pupil along the axis of that quadrant.
3. **Hold the condensing lens with the more convex surface toward the patient**, about **2 inches** (roughly the focal length) from the eye, and align lens, pupil, and your line of sight.
4. **Find the image with the lens close, then withdraw slowly** until the image fills the lens. Small movements toward and away from the eye change magnification and field; small lateral tilts kill the image.
5. **Move the lens, not your head**, for fine adjustment.
6. **If you lose the view**, do not freeze. Reduce the working distance, re-find the pupil, then withdraw. Ask "Do you have a view?" while performing it, and troubleshoot for anything obstructing the central camera.
7. **Remember the image is inverted and reversed.** When the patient looks up and you view from below, you are seeing the superior retina, and it appears inverted in the lens. Name the *anatomic* quadrant, not what appears at the top of your view.
8. **Look for the vortex vein ampulla deliberately.** It is the defined marker that your view is genuinely peripheral, and naming it demonstrates the view meets the criterion.

Skill 5 • Dilated biomicroscopy and non-contact fundus lens (items 22 to 43)

This skill has two halves. **Items 23 to 28 are performed with the biomicroscope alone**, no fundus lens: the crystalline lens (direct and retroillumination) and the retrolental space and anterior vitreous. **Items 29 to 42 are performed with the non-contact fundus lens**: posterior vitreous, optic nerve, arcades, and macula.

PART A: BIOMICROSCOPE ALONE (ITEMS 22 TO 28)

Item	What is scored	Technique and words
22	Explain the purpose to the patient	<i>"Now I am going to use the microscope to look at the lens inside your eye and the jelly behind it, and then I will hold a lens in front of your eye to look at the optic nerve and the central retina. Nothing will touch your eye. The light is bright, and you can blink whenever you need to."</i>
23	Properly scan all layers of the crystalline lens	Parallelepiped or optic section, 16× , angle 30 to 45 degrees . Focus progressively through every layer and say each one: <i>"I am scanning through all layers of the crystalline lens: anterior capsule, anterior cortex, the adult, infantile, fetal and embryonic nuclei, posterior cortex, and posterior capsule."</i> Move the focus in a controlled anteroposterior sweep, not a single plane.

Item	What is scored	Technique and words
24	Accurately describe the crystalline lens	<i>"The crystalline lens is clear throughout all layers, with a clear anterior and posterior capsule, no nuclear sclerosis, no cortical spoking, no posterior subcapsular changes, no vacuoles, and no pseudoexfoliative material on the anterior capsule." If cataract is present, grade it by type and severity, for example "grade 2 nuclear sclerosis with trace cortical spoking inferonasally."</i>
25	Properly examine the lens using retroillumination · maximum illumination for this item only, and you must move the light source	Turn the illumination to maximum , use a moderate beam, and displace the illumination arm so the beam passes through the pupil to the retina and returns , backlighting the lens against the fundus reflex. Move the light source across the pupil to retro-illuminate the whole lens. Announce all of it: <i>"I am now using retroillumination, with the illumination at maximum, and moving the light source across the pupil to backlight the lens against the fundus reflex."</i>
26	Accurately describe the lens on retroillumination	<i>"On retroillumination the lens is clear with an even, unbroken fundus reflex. There are no cortical spokes, no posterior subcapsular opacities, no vacuoles, and no lens opacities silhouetted against the reflex."</i>
27	Properly examine the anterior vitreous	Return to normal illumination, focus posterior to the lens into the retrolental space and anterior vitreous , with a conical or narrow beam and higher illumination, in a darkened room. <i>"I am now focusing behind the lens into the retrolental space and anterior vitreous."</i>
28	Accurately describe the anterior vitreous	<i>"The retrolental space and anterior vitreous are optically clear with no cells, no pigment granules, no haemorrhage, no asteroid hyalosis, and no anterior vitreous strands."</i>

PART B: WITH THE NON-CONTACT FUNDUS LENS (ITEMS 29 TO 43)

Item	What is scored	Technique and words
29	Provide appropriate instructions for fixation	<i>"Please keep your chin on the rest and your forehead forward, keep both eyes open, and look straight ahead over my shoulder at the corner of the wall. Keep looking at that spot even though the light is bright, and blink whenever you need to. If I need you to look in a different direction I will tell you."</i>
30	Properly position the fundus lens: alignment, centration, distance	Pull the slit lamp back, set a moderate beam and low illumination initially, then hold the lens about 7 to 10 mm in front of the cornea, centred on the pupil and perpendicular to the visual axis . Advance the slit lamp until the fundus image appears, then fine-focus. Steady your hand on the patient's forehead or the headrest without pressing on the eye.
31	Properly examine the posterior vitreous	Focus just anterior to the retinal surface. <i>"I am now examining the posterior vitreous."</i> Use a slightly wider beam and higher illumination, and scan.
32	Accurately describe the posterior vitreous	<i>"The posterior vitreous is clear with no cells, no pigment, no haemorrhage, and no posterior vitreous detachment: there is no Weiss ring and no separation of the posterior hyaloid." If a PVD is present, say so and name the Weiss ring.</i>

Item	What is scored	Technique and words
33	Properly examine the optic nerve	Centre the disc, use a beam narrower than the disc for contour assessment, and scan the whole disc and the peripapillary area. Announce: <i>"I am now examining the optic nerve head."</i>
34	Accurately describe the optic nerve: colour, rim integrity, contour, PPA	All four named features must be stated. <i>"The optic nerve head is pink and healthy in colour with no pallor. The neuroretinal rim is intact throughout with no notching or focal thinning, and rim thickness follows the ISNT rule with inferior thicker than superior, thicker than nasal, thicker than temporal. The contour of the cup is regular with a gradual slope and no undermining or laminar dot sign. There is no peripapillary atrophy and no disc haemorrhage."</i>
35	Accurately state the horizontal cup-to-disc ratio	State it as a separate, explicit number. <i>"The horizontal cup-to-disc ratio is 0.3."</i>
36	Accurately state the vertical cup-to-disc ratio	<i>"The vertical cup-to-disc ratio is 0.3."</i> Two separate items: state two separate numbers.
37	Properly examine and obtain a clear image of the inferior and superior temporal arcades	Follow each arcade from the disc out toward the periphery. Ask the patient to look slightly temporally if needed. <i>"I am now following the superior temporal arcade out from the disc, and now the inferior temporal arcade."</i>
38	Accurately identify and assess the temporal arcades	<i>"The superior and inferior temporal arcades are of normal calibre and course, with a normal arteriole-to-venule ratio of approximately two to three, no arteriovenous nicking, no copper or silver wiring, no tortuosity, no sheathing, no emboli, no haemorrhage, no exudate, and no cotton wool spots along either arcade."</i>
39	Properly examine and obtain a clear image of the inferior and superior nasal arcades	Ask the patient to look slightly nasally. <i>"Please look a little toward your nose. I am now following the superior nasal arcade, and now the inferior nasal arcade."</i>
40	Accurately identify and assess the nasal arcades	<i>"The superior and inferior nasal arcades are of normal calibre and course with a normal arteriole-to-venule ratio, no nicking, no haemorrhage, no exudate, and no cotton wool spots."</i>
41	Properly examine both the fovea and the macula	Two structures, both required. Narrow the beam and reduce illumination for patient comfort, and warn: <i>"This next part is the brightest, and it will only take a few seconds. Please keep looking straight at my shoulder. I am now examining the macula, and now the fovea."</i>
42	Accurately describe both the fovea and the macula	Describe them separately. <i>"The macula is flat and intact with even pigmentation, no drusen, no pigmentary mottling, no haemorrhage, no exudate, and no oedema or cystic change. The fovea is flat with a crisp foveal reflex present and no atrophy, no pigment change, and no lamellar or full-thickness defect."</i>

Item	What is scored	Technique and words
43	Smooth and logical sequence	Lens layers, lens retroillumination, anterior vitreous, then the fundus lens: posterior vitreous, optic nerve with both cup ratios, temporal arcades, nasal arcades, then macula and fovea last (because it is the brightest and least comfortable).
44	Maintain proper hygiene throughout the station	Wash hands during observation, then do not touch your face or hair. Disinfect the headrest and chin rest and the fundus lens. If you contaminate your hands, wash them again before touching the patient or the contact surfaces.

THE TEN ITEMS MOST COMMONLY LOST IN THIS STATION

1. **Peripheral views with the arcades in view.** If you can see the arcades, that is not a peripheral view and the item is not earned. Push out further until a vortex ampulla appears.
2. **Not naming the vortex vein ampulla**, which is the defined marker of a genuine peripheral view.
3. **Naming the wrong quadrant**, because the image is inverted and reversed. Name the anatomic quadrant based on where the patient is looking and where you are standing.
4. **Adjusting the BIO rheostat**, which the guide explicitly tells you not to do.
5. **Item 25:** performing retroillumination without turning the illumination to **maximum** and without **moving the light source**.
6. **Item 23:** focusing on the lens in one plane instead of scanning through all layers and naming them.
7. **Items 35 and 36:** giving one combined cup-to-disc ratio instead of two separate numbers.
8. **Item 34:** describing the disc without covering all four required features (colour, rim integrity, contour, and peripapillary atrophy).
9. **Items 39 and 40:** forgetting the **nasal** arcades entirely after doing the temporal ones.
10. **Items 41 and 42:** describing the macula but not the **fovea** separately.

A FIFTEEN-MINUTE PACING PLAN FOR THE POSTERIOR STATION

Clock	Task
0:00–0:40	Introduce, ask which eye, explain BIO, obtain positioning consent, confirm the patient is comfortable.
0:40–7:00	BIO. Eight quadrants in one continuous rotation, roughly 40 seconds each, naming the quadrant, the vortex ampulla, and the assessment. Then the posterior pole.
7:00–9:30	Biomicroscope alone. Explain, scan all lens layers, retroillumination at maximum with the light source moving, anterior vitreous.
9:30–14:30	Fundus lens. Fixation instructions, lens positioning, posterior vitreous, optic nerve with both cup ratios, temporal arcades, nasal arcades, macula and fovea last.
14:30–15:00	Buffer. Remain in the room until the announcement.

Forty seconds per quadrant is the discipline that makes this station work. Practise with a timer and a partner calling out the elapsed time until eight quadrants in six and a half minutes feels unhurried.

RETRIEVAL PRACTICE

1. Give the exam's definition of a peripheral view.
2. What must *not* be in view during the eight BIO peripheral items, and what must be?
3. Where do you stand and where does the patient look to view the inferior peripheral retina?

4. What is the optimal BIO working distance, and what must you not adjust?
5. How many quadrants are scored in BIO, and how many items does each generate?
6. Which item requires maximum illumination, and what else does it require?
7. Name the lens layers you should verbalise when scanning the crystalline lens.
8. Which items are performed *without* the fundus lens?
9. What four features must you state when describing the optic nerve?
10. How many cup-to-disc ratios must you state, and how?
11. Which arcades are frequently forgotten, and how do you position the patient for them?
12. Why should the macula and fovea be examined last?
13. You lose the BIO view. Give three troubleshooting steps.
14. Why can naming a quadrant go wrong, and how do you avoid it?
15. What is your time budget per BIO quadrant?

KEY

1. An area of retina **extending from the equator to the ora serrata** and containing **at least a partial view of a vortex vein ampulla**.
2. **The arcades must not be in view**; the condensing lens should be filled as much as possible, and a **vortex vein ampulla should be visible**.
3. Stand at the patient's **superior** side, above the head; the patient looks **down**.
4. **18 to 25 inches**; **do not adjust the BIO rheostat**, which the SP has set to its calibrated level.
5. **Eight** peripheral quadrants, each generating **two items**: one for obtaining the view and one for identifying the quadrant and giving an assessment, plus two more items for the posterior pole.
6. **Item 25**, lens retroillumination; it also requires that you **move the light source** to obtain the retroilluminated image.
7. Anterior capsule, anterior cortex, adult nucleus, infantile nucleus, fetal nucleus, embryonic nucleus, posterior cortex, and posterior capsule.
8. **Items 23 to 28**: crystalline lens by direct illumination, crystalline lens by retroillumination, and the retrolental space and anterior vitreous.
9. **Colour, rim integrity, contour, and peripapillary atrophy**.
10. **Two**, as separate explicit numbers: the horizontal ratio and the vertical ratio, each its own item.
11. The **nasal** arcades; ask the patient to look slightly toward the nose.
12. Because it is the brightest and least comfortable part, so doing it last preserves patient cooperation for everything else.
13. Move the lens closer to re-find the pupil then withdraw slowly; keep the lens perpendicular and centred and move the lens rather than your head; ask *"Do you have a view?"* while performing it and check nothing is obstructing the central camera.
14. Because the BIO image is **inverted and reversed**. Name the **anatomic** quadrant based on where the patient is looking and where you are standing, not on where the feature appears within the lens.
15. About **40 seconds**, giving eight quadrants plus the posterior pole in roughly six and a half minutes.

CHAPTER RECAP

1. A peripheral view runs from the equator to the ora and contains a vortex vein ampulla; name the ampulla.
2. No arcades in the eight peripheral items; fill the lens as much as possible.
3. Stand opposite the quadrant, patient looks toward it, and name the anatomic quadrant despite the inverted image.
4. Work distance 18 to 25 inches, and never touch the rheostat.
5. Forty seconds per quadrant, one continuous rotation, then the posterior pole.
6. Items 23 to 28 use the biomicroscope alone; retroillumination needs maximum illumination and a moving light source.
7. Scan and name all lens layers, not one plane.
8. The optic nerve needs colour, rim, contour, and PPA, then two separate cup-to-disc numbers.
9. Do the nasal arcades as well as the temporal, and describe the fovea separately from the macula.
10. Macula and fovea last, and hygiene throughout.

Findings lexicon: the exact words to say

Half of the skills items are "accurately describe"

Never say WNL, fine, OK, or not bad

THE RULE THIS CHAPTER EXISTS TO ENFORCE

The candidate guide is explicit: **state your findings in the same manner as you would enter them into a patient's medical record**, and **avoid terminology such as "OK", "fine", "WNL", or "not bad"**. Every "accurately describe" item is earned by naming the structure and then describing its state in clinical terms.

The pattern to internalise for every structure is: **name the structure, state that you are examining it, then give a positive description plus the pertinent negatives**. Positive description first, negatives second, and never a bare "normal".

Anterior segment lexicon

STRUCTURE BY STRUCTURE, WITH A NORMAL SCRIPT AND THE ABNORMAL VOCABULARY

Structure	Normal script	Abnormal vocabulary you may need
Lids and lashes	"Lid skin intact, no lesions, oedema, or erythema. Lashes normal in number, direction, and distribution."	Erythema, oedema, ecchymosis, laceration, chalazion, hordeolum, papilloma, seborrheic keratosis, basal cell carcinoma with a pearly rolled edge, ptosis with MRD1 measurement, dermatochalasis, ectropion, entropion, lagophthalmos, madarosis, poliosis, trichiasis, distichiasis, collarettes (cylindrical dandruff), scurf, crusting.
Lid margins	"Margins smooth and regular, no notching, telangiectasia, or thickening. Meibomian orifices patent and non-capped."	Capped or plugged orifices, pouting, telangiectasia, thickening, keratinisation, notching, posterior displacement of the mucocutaneous junction, turbid or toothpaste-like expression, gland dropout, foamy tear film, anterior or posterior blepharitis.
Puncta	"Lower punctum patent, normal size, correctly apposed to the globe."	Stenotic, occluded, everted, agenesis, punctal plug in situ, pouting with expressible concretions (canaliculitis), supernumerary punctum.
Palpebral conjunctiva	"Pink and smooth, no follicles, papillae, concretions, membranes, or scarring. Fornix of normal depth."	Follicles (avascular with encircling vessels), papillae with a central vessel, giant papillae over 0.3 mm, membrane, pseudomembrane, concretions, symblepharon, fornix foreshortening, Arlt line, subtarsal foreign body, chemosis, injection graded 1 to 4.

Structure	Normal script	Abnormal vocabulary you may need
Bulbar conjunctiva and sclera	"Conjunctiva clear and quiet, no injection or chemosis. Limbus intact with clear palisades of Vogt. Sclera white, no thinning or nodules."	Diffuse or sectoral injection, ciliary flush, chemosis, subconjunctival haemorrhage, pinguecula, pterygium with a Stocker line, follicles, papillae, conjunctivochalasis, pigmented lesion with or without intralesional cysts, primary acquired melanosis, salmon-pink infiltrate, episcleral versus deep vascular injection, scleral nodule, scleral thinning with uveal show, staphyloma.
Cornea, epithelium	"Epithelium intact, smooth, and clear, no punctate staining, defect, or oedema."	Superficial punctate keratitis, graded and located; epithelial defect with measured dimensions; dendrite with terminal bulbs; pseudodendrite; microcystic oedema; bullae; map-dot-fingerprint changes; Salzmann nodules; iron line (Hudson-Stähli, Fleischer, Stocker, Ferry); verticillata; recurrent erosion.
Cornea, stroma	"Stroma clear, no infiltrate, scar, neovascularisation, thinning, or striae."	Infiltrate with measured size, depth, location, and margin character (feathery, ring, satellite); scar; oedema with striae or folds; thinning with a measured percentage; neovascularisation, superficial or deep, with the clock hours; Vogt striae; lipid deposition; crystalline deposits; Wessely immune ring; Descemet folds.
Cornea, endothelium and Descemet	"Descemet membrane and endothelium clear, no guttae, keratic precipitates, or pigment."	Central guttae with a beaten-metal appearance; peripheral Hassall-Henle bodies; keratic precipitates, fine and non-pigmented or large and mutton-fat, in an Arlt triangle; Krukenberg spindle; pigment dusting; Khodadoust line; break in Descemet with hydrops; Kayser-Fleischer ring.
Anterior chamber depth (Van Herick)	"Temporal shadow depth greater than one half of the corneal section thickness: Van Herick grade 4, wide open."	Grade 3 (between a quarter and a half), grade 2 (one quarter), grade 1 (less than one quarter, occludable); shallow peripherally with a convex iris; irregularly shallow.
Anterior chamber contents	"Grade zero cells, grade zero flare, optically clear aqueous."	Cells graded 0.5+ through 4+ by SUN; flare graded 1+ through 4+; fibrin; hypopyon with a measured height; hyphaema graded by the fraction filled; pigment; white blood cells versus red; plasmoid aqueous.
Iris	"Iris flat, normal architecture and colour, intact collarette, no naevi, nodules, neovascularisation, atrophy, or synechiae. Pupil round, central, normal size, with an intact pupillary ruff."	Naevus, Lisch nodules, Koeppe and Busacca nodules, neovascularisation at the pupil margin or in the angle, sectoral atrophy, diffuse stromal atrophy with heterochromia, iridodialysis, sphincter tears, corectopia, polycoria, transillumination defects (mid-peripheral radial or peripupillary moth-eaten), posterior or anterior synechiae, iris bombé, plateau configuration, persistent pupillary membrane.

Structure	Normal script	Abnormal vocabulary you may need
Iris transillumination	"On transillumination the entire iris is intact with no transillumination defects."	Mid-peripheral radial spoke-like defects (pigment dispersion), peripupillary moth-eaten defects (pseudoexfoliation), sectoral defects (post-herpetic or post-iridotomy), full-thickness defect from trauma or a laser iridotomy.
Crystalline lens	"Lens clear throughout all layers, clear anterior and posterior capsule, no nuclear sclerosis, cortical spoking, or posterior subcapsular change."	Nuclear sclerosis graded 1 to 4; cortical spoking with location; posterior subcapsular plaque; anterior subcapsular; polar opacity; vacuoles; pseudoexfoliative material in a target pattern; phacodonesis; subluxation with the direction; Vossius ring; rosette cataract; Christmas tree; oil droplet; sunflower; Mittendorf dot; posterior capsular opacification; Elschnig pearls; an intraocular lens with its position and centration.
Anterior vitreous	"Retrolental space and anterior vitreous optically clear, no cells, pigment, haemorrhage, or asteroid hyalosis."	Cells graded, pigment granules (tobacco dust or Shafer sign), red blood cells, asteroid hyalosis, anterior vitreous strands, vitreous in the anterior chamber, synergetic cavities.

Angle lexicon

GONIOSCOPIC DESCRIPTION

Element	How to state it
Quadrant	Name the anatomic quadrant and the mirror: "I am viewing the inferior angle through the superior mirror."
Most posterior visible structure	Name it precisely: Schwalbe line, anterior or posterior trabecular meshwork, scleral spur, ciliary body band, or iris root. This is a scored element in every quadrant.
Angular width	Shaffer grade: grade 4 is 35 to 45 degrees and wide open; grade 3 is 20 to 35 ; grade 2 is 20 and narrow; grade 1 is 10 and extremely narrow; grade 0 is closed.
Iris approach and configuration	"Regular and flat with a normal insertion", or "convex and bowed forward", or "concave", or "plateau configuration with a double hump on indentation". Required explicitly for the inferior quadrant.
Pigmentation	Graded 0 to 4: "trabecular pigmentation grade 1", "grade 3 homogeneous dense pigmentation", "Sampaolesi line present".
Abnormalities	Peripheral anterior synechiae (broad, tent-like, obscuring underlying structures) versus iris processes (fine, lace-like, not obscuring); neovascularisation; blood in Schlemm canal; angle recession with an abnormally wide ciliary body band; foreign body; cyclodialysis cleft; posterior embryotoxon.

Posterior segment lexicon

STRUCTURE BY STRUCTURE

Structure	Normal script	Abnormal vocabulary
Posterior vitreous	"Posterior vitreous clear; no cells, pigment, or haemorrhage, and no posterior vitreous detachment: no Weiss ring."	Posterior vitreous detachment with a Weiss ring; vitreous haemorrhage; pigment granules; vitreous cells graded; snowballs; vitreous condensation; vitreomacular adhesion; asteroid hyalosis.
Optic nerve, colour	"Pink and healthy in colour with no pallor."	Temporal pallor; diffuse pallor; hyperaemia, swelling with obscured margins, chalky white pallid swelling.
Optic nerve, rim integrity	"Neuroretinal rim intact throughout with no notching or focal thinning; rim thickness follows the ISNT rule."	Inferior or superior notching; focal or diffuse rim thinning; ISNT violation; rim pallor; acquired pit of the optic nerve.
Optic nerve, contour	"Cup contour regular with a gradual slope, no undermining and no laminar dot sign."	Steep or undermined cup walls; laminar dot sign; bayoneting of vessels; nasalisation; vertical cup elongation; tilted disc with situs inversus; crowded small disc; myelinated nerve fibres; optic disc drusen with a lumpy margin; double ring sign.
Optic nerve, PPA	"No peripapillary atrophy and no disc haemorrhage."	Alpha and beta zone peripapillary atrophy; peripapillary crescent; disc haemorrhage at a stated clock hour; peripapillary staphyloma; choroidal neovascularisation at the disc margin.
Cup-to-disc ratio	"Horizontal cup-to-disc ratio 0.3. Vertical cup-to-disc ratio 0.3."	State two separate numbers; note asymmetry between eyes if relevant; comment on disc size, because a small disc makes a normal cup look large and a large disc conceals rim loss.
Arcades and vessels	"Arcades of normal calibre and course, arteriole-to-venule ratio approximately two to three, no nicking, wiring, tortuosity, sheathing, emboli, haemorrhage, exudate, or cotton wool spots."	Attenuation; arteriovenous nicking; copper or silver wiring; tortuosity; venous beading; sheathing; boxcarring; Hollenhorst plaque or calcific embolus; intraretinal microvascular abnormalities; neovascularisation of the disc or elsewhere; collateral vessels; sclerosis.
Macula	"Macula flat and intact with even pigmentation, no drusen, pigmentary mottling, haemorrhage, exudate, or oedema."	Hard and soft drusen; reticular pseudodrusen; pigmentary mottling; geographic atrophy with measured extent; subretinal fluid; subretinal haemorrhage; a greyish-green subretinal lesion; disciform scar; cystoid change; epiretinal membrane with retinal striae; hard exudate in a circinate ring; a macular star.
Fovea	"Fovea flat with a crisp foveal reflex present, no atrophy, pigment change, or lamellar or full-thickness defect."	Blunted or absent foveal reflex; full-thickness macular hole with a stated size; lamellar hole; pseudohole; cherry red spot; foveal hypoplasia; a yellow vitelliform lesion; foveal cyst.

Structure	Normal script	Abnormal vocabulary
Peripheral retina, per quadrant	"[Quadrant] peripheral retina from the equator to the ora serrata is flat and intact, vortex vein ampulla visible, with no holes, tears, lattice degeneration, pigmentary change, or retinal detachment."	Horseshoe (flap) tear with the clock hour; operculated hole; atrophic round hole; retinal dialysis; lattice degeneration with the clock hours and whether it contains holes; pavingstone (cobblestone) degeneration; peripheral cystoid degeneration; retinoschisis; white with or without pressure; meridional folds; enclosed oral bays; snail track degeneration; chorioretinal scar; pigmented demarcation line; sea fan neovascularisation; subretinal fluid.
Background retina	"Background retina flat with no haemorrhage, exudate, or pigmentary change."	Dot-blot haemorrhage; flame-shaped haemorrhage; microaneurysm; cotton wool spot; hard exudate; Roth spot; retinal whitening; commotio; choroidal rupture; drusen; chorioretinal atrophy; myelinated nerve fibres; naevus with measured dimensions; laser scars.

Phrases to use, and phrases never to use

LANGUAGE DISCIPLINE

Never say	Say instead
"Lids are fine."	"Lid skin is intact with no lesions, oedema, or erythema, and the lashes are normal in number, direction, and distribution."
"Cornea WNL."	"The epithelium is intact and clear with no staining, the stroma is clear with no infiltrate or scar, and the endothelium is clear with no guttae or precipitates."
"Anterior chamber deep and quiet."	"Van Herick grade 4 temporally, and grade zero cells with grade zero flare in an optically clear aqueous."
"Angle open."	"Inferior angle viewed through the superior mirror: the most posterior visible structure is the ciliary body band, the iris approach is regular and flat, Shaffer grade 4, trabecular pigmentation grade 1, no synechiae."
"Pressure 16."	"The intraocular pressure of the right eye by Goldmann applanation is 16 millimetres of mercury at 10:42."
"Disc looks good."	"The optic nerve head is pink with no pallor, the rim is intact and follows the ISNT rule, the contour is regular with no undermining, there is no peripapillary atrophy and no disc haemorrhage. Horizontal cup-to-disc 0.3, vertical cup-to-disc 0.3."
"Macula OK."	"The macula is flat and intact with even pigmentation and no drusen, mottling, haemorrhage, exudate, or oedema. The fovea is flat with a crisp foveal reflex."
"Periphery clear."	"The superior-temporal peripheral retina from the equator to the ora serrata is flat and intact with a vortex vein ampulla in view, and there are no holes, tears, or lattice degeneration."
"Not bad."	Nothing. Describe it.

Never say	Say instead
"Everything looks normal."	Name each structure and describe it individually. A global statement earns no individual items.

TWO HABITS THAT GENERATE THE LANGUAGE AUTOMATICALLY

1. **Announce, then describe.** "I am now examining X" followed by "X is ..." creates the two-part structure the form rewards, because most structures generate a *properly examine* item and an *accurately describe* item.
2. **Two positives and three negatives.** For every structure, give two positive descriptors (colour, contour, clarity, calibre, integrity) and three pertinent negatives (the abnormalities that would matter most for that structure). That formula produces a record-quality description in under ten seconds and prevents the reflex to say "normal".

RETRIEVAL PRACTICE: CONVERT EACH TO RECORD LANGUAGE, OUT LOUD

1. "Lids and lashes normal."
2. "Upper palpebral conjunctiva clear."
3. "Cornea clear."
4. "Chamber deep and quiet."
5. "Iris normal."
6. "Transillumination normal."
7. "Angle open all round."
8. "Lens clear."
9. "Pressure normal."
10. "Disc healthy, cup 0.3."
11. "Vessels normal."
12. "Macula and fovea normal."
13. "Periphery flat, no holes."
14. "Vitreous clear."

KEY: MODEL PHRASING

1. "Lid skin intact, no lesions, oedema, or erythema; lashes normal in number, direction, and distribution, with no collarettes, madarosis, or trichiasis."
2. "Superior palpebral conjunctiva pink and smooth on eversion, no follicles, no papillae greater than 0.3 millimetres, no concretions, no membranes, no scarring, and no subtarsal foreign body."
3. "Epithelium intact and clear with no punctate staining or defect; stroma clear with no infiltrate, scar, thinning, or neovascularisation; Descemet and endothelium clear with no guttae or keratic precipitates."
4. "Van Herick grade 4 temporally; grade zero cells and grade zero flare in an optically clear aqueous."
5. "Iris flat with normal architecture and colour, intact collarette, no naevi or nodules, no neovascularisation, no atrophy, and no synechiae; pupil round, central, and of normal size with an intact pupillary ruff."
6. "On transillumination the entire iris is intact, with no mid-peripheral radial and no peripupillary transillumination defects."
7. "All four quadrants: most posterior visible structure is the ciliary body band inferiorly, nasally, and temporally, and the anterior trabecular meshwork superiorly; iris approach regular and flat; Shaffer grade 4 inferiorly, nasally, and temporally and grade 2 to 3 superiorly; pigmentation grade 1; no peripheral anterior synechiae."
8. "Crystalline lens clear through all layers, clear anterior and posterior capsule, no nuclear sclerosis, cortical spoking, or posterior subcapsular change; on retroillumination the fundus reflex is even and unbroken."
9. "Intraocular pressure right eye by Goldmann applanation 16 millimetres of mercury at 10:42."
10. "Optic nerve pink with no pallor; neuroretinal rim intact throughout following the ISNT rule with no notching; cup contour regular with no undermining

or laminar dot sign; no peripapillary atrophy and no disc haemorrhage. Horizontal cup-to-disc ratio 0.3; vertical cup-to-disc ratio 0.3."

11. "Superior and inferior temporal and nasal arcades of normal calibre and course, arteriole-to-venule ratio approximately two to three, no nicking, wiring, tortuosity, emboli, haemorrhage, exudate, or cotton wool spots."
12. "Macula flat and intact with even pigmentation, no drusen, mottling, haemorrhage, exudate, or oedema; fovea flat with a crisp foveal reflex present."

13. "[Named quadrant] peripheral retina from the equator to the ora serrata is flat and intact with a vortex vein ampulla in view; no holes, tears, lattice degeneration, pigmentary change, or subretinal fluid."

14. "Anterior vitreous and retrolental space optically clear with no cells, pigment, haemorrhage, or asteroid hyalosis; posterior vitreous clear with no posterior vitreous detachment and no Weiss ring."

CHAPTER RECAP

1. Half the skills items are description items, earned only by speaking in record language.
2. Never say WNL, fine, OK, or not bad. Describe.
3. Announce the structure, then describe it: that mirrors the two-item structure of the form.
4. Two positive descriptors plus three pertinent negatives, every time.
5. Grade what can be graded: Van Herick, SUN cells and flare, Shaffer, pigmentation, cup ratios.
6. State the pressure with eye, method, and time.
7. Name the quadrant, the most posterior angle structure, and the iris approach in gonioscopy.
8. Give the optic nerve four features and two separate cup ratios.
9. Describe the fovea separately from the macula.
10. Name the vortex vein ampulla in every peripheral view.

Views, repeats, stops, hygiene, and safety

- Four attempts then a stop
- Repeat erases all prior scores for that skill
- Hygiene is a scored item in both stations

ANSWER THESE COLD BEFORE YOU SIT THE EXAM

1. What does the examiner confirming a view tell you, and what does it not?
2. How many attempts do you get at an item, and what happens after that?
3. What is the difference between repeating an item and repeating a skill?
4. What happens to your prior scores if you announce a skill repeat?
5. What happens after a stop?
6. Name five ways to break hygiene protocol.

ATTEMPTS

4
per item, after which you are stopped and all remaining items in that skill are scored "no".

Views: obtaining, confirming, and troubleshooting

THE RULES ON VIEWS, PRECISELY AS THE GUIDE STATES THEM

Rule	Detail
Cameras	The slit lamp camera is mounted on the left ocular ; you are scored on what the camera captures through the left ocular, so ensure the image is fully seen there. The BIO camera is mounted centrally , with images recorded through each ocular. Cameras, monitors, and recordings are calibrated, so if the view is not obstructed, the recorded image is exactly what you see.
You are responsible for the view	It is your responsibility to ensure the examiner or proctor has a view. Nobody will volunteer that they do not.
Asking	You may ask at any point <i>while performing the view</i> . You will be told either "I have a view" or "I do not have a view" .
What confirmation means	Only that something is visible . It does not tell you the quality of the view, whether it meets the minimum criteria for the item, or whether it is of the correct structure, angle, or quadrant.
Asking afterwards	Useless. You will be told "I can only comment on a view while it is being performed."
If told there is no view	Troubleshoot, and specifically ensure nothing obstructs the left ocular of the slit lamp or the central camera of the BIO headset : your hand, the lens edge, a lid, lashes, the condensing lens rim, or your own head position.
Monitors	There is a monitor in the room for the examiner or proctor only. You may not look at it at any time; attempting to do so may be treated as improper conduct.
Fleeting views	Do not count. Hold the view long enough for the examiner to see and score it, and while you hold it, describe it.
"Entire"	Means 360 degrees or the complete structure . Sweep and say that you are sweeping.

A TROUBLESHOOTING SEQUENCE TO RUN AUTOMATICALLY

- 1 **Ask, while performing:** *"Do you have a view?"* Do it early rather than after four failed attempts.
- 2 **Check for obstruction.** Move your hand, fingers, and the lens rim out of the optical path; move your head so you are not blocking the ocular; check the patient's lashes and lids are not in the way.
- 3 **Reset the basics.** Confirm the patient's chin and forehead are in position, confirm gaze direction, and confirm your own focus and illumination.
- 4 **Change one variable at a time.** With the fundus lens or condensing lens, change the lens-to-eye distance first, then the slit lamp position, then the beam width. Changing three things at once loses the reference point.
- 5 **Re-find and re-withdraw.** With BIO or the fundus lens, move closer until the image appears, then withdraw slowly until it fills the lens.
- 6 **Re-instruct the patient.** Most lost views are a gaze problem, not an optics problem. *"Look further up for me, all the way, as if you are trying to see your own eyebrows."*
- 7 **Move on if you have used two attempts.** An item is worth one point; the remaining items are worth many. **You will be stopped after four attempts, and a stop scores every remaining item in that skill as "no".**

Repeats

REPEATING AN ITEM VERSUS REPEATING A SKILL

Situation	Rule and consequence
Repeating an item	Permitted within the same skill . The last intentional performance is what is scored . You do not need to announce an item repeat, though saying "I would like to repeat that" removes ambiguity. This is generally safe and worth doing for a single item you know you performed poorly, provided you have time.
Repeating a skill	You must announce your intention before you begin . The moment you announce it, all previously recorded scores for that skill are erased and you are evaluated as if performing the skill for the first time . That means you must redo <i>every</i> item in the skill, and the time cost is usually prohibitive.
A stopped procedure or skill	Cannot be repeated.
Where the policy is posted	On the door of each skills room. Read it during observation time.

THE REPEAT TRAP, STATED PLAINLY

The candidate guide contains two explicit cautions, and both come from observed candidate behaviour:

- **Candidates who have not watched the clock and then try to repeat an entire skill for one missed item frequently run out of time**, losing far more than the single item they were chasing.
- **Candidates repeating a skill often focus so heavily on the previously missed item that they overlook others**, and end with a *lower* score than they had already earned.

The operational rule: repeat an *item* freely if you have time. Repeat a *skill* almost never, and only if you are early in the skill, you have made a fundamental error such as working on the wrong eye or omitting a sequence of items, and you have clearly enough time to complete every item again.

Attempts, interventions, and stops

WHAT HAPPENS WHEN THINGS GO WRONG

Event	Rule
Attempts per item	You are allowed four attempts . On unsuccessfully completing the fourth, the examiner will stop you and instruct you to move to the next skill.
Consequence of a stop	Every remaining item in that skill is scored "no", and the skill cannot be repeated. This is why you must not burn four attempts on one difficult quadrant.
Safety stop	If you endanger the standardised patient you will be stopped immediately . The SP may also stop you if they are concerned about their own safety.
Intervention	If the SP, proctor, or examiner intervenes, corrective action is expected before you proceed . Do not carry on as though nothing happened: acknowledge it, correct it out loud, then continue. For example, if you are asked to wash your hands, wash them and say that you are doing so.
Sequencing	Items are listed in the order in which they should optimally be performed, but you may alter the sequence of items within a skill as long as your sequence makes logical sense , and you may alter the order of skills within a station .
Declaring you are finished	Saying "I am finished with this station" ends scoring. The examiner leaves, the SP stays and begins preparing the room, and no further items are scored . It is your choice, and there is rarely a good reason to do it. If you make a confusing statement or start casual conversation, you will be reminded that it is your responsibility to state whether you are finished.
Equipment failure	Tell the SP, who summons staff. Staff resolve it and instruct you how to proceed, including granting additional time if appropriate. Additional time appears in the top left corner of your device . Non-affected candidates are paused at the end of the cycle so everyone resynchronises.
Injury to you	Notify staff immediately. Cover any open wound. Use universal precautions. Any bleeding must be reported and stopped before continuing. No additional time is given for an injury caused by candidate error.

Hygiene and patient safety

HYGIENE IS A SCORED ITEM IN BOTH STATIONS. HERE IS HOW IT IS LOST.

1. **Touching your face or hair after washing your hands.** This is named explicitly in the candidate guide as breaking protocol, and you may then be required to wash again before touching the patient or the contact surfaces of the equipment. It is the single commonest hygiene failure, and it happens when candidates push up their glasses or move their hair.
2. **Not washing at all**, because you assumed disinfecting the slit lamp was sufficient. Wash to CDC standard during observation time.
3. **Failing to state that you would disinfect the instruments** (anterior station item 1).
4. **Touching the eye or lashes with a bottle tip** when instilling a drop.
5. **Contaminating the gonioscopy lens or the fundus lens** by setting it down on a non-clean surface, or handling the optical face.
6. **Not covering an open wound** on a finger or hand.
7. **Reusing a cotton-tipped applicator** or a tissue between structures or between uses.

Practical prevention: tie your hair back for the photograph and the exam; push your glasses up *before* you wash; wash during observation and then keep your hands in your field of view; and if you do touch your face, **say so and wash again**. Correcting it out loud demonstrates the behaviour the item is assessing.

PATIENT SAFETY BEHAVIOURS THAT ARE SCORED OR THAT PREVENT A STOP

Behaviour	What to do
Warn before every touch	"I am going to turn your eyelid out now." "The lens is going to touch your eye now; it will feel like a slight pressure."
Warn before every bright light	"This next light is the brightest one and it will be uncomfortable for a few seconds."
Respect personal space	Do not lean on the patient, do not brace on their face harder than necessary, and do not touch below the shoulder .
No handshake	Not permitted, for hygiene.
Light exposure discipline	Do not hold the beam on one area for an extended period, particularly the fovea. Reduce illumination when you can, and examine the macula last.
Accept blinks and breaks	The SP may blink or ask for a break. Grant it, say something reassuring, and resume.
Instrument control	Never release the gonioscopy lens while it is on the eye; retract the tonometer smoothly; do not advance the slit lamp into the patient.
Drop technique	Do not touch the eye or lashes with the bottle; instil into the correct eye and say which; ask the patient to close gently rather than squeeze.
Check comfort	Ask once or twice during the station: "Are you comfortable? Let me know if you need a break."

Communication and professionalism, which the SP scores

THE BEHAVIOURS THAT EARN THE 11% COMMUNICATION AND PROFESSIONALISM DOMAIN

Domain element	What it looks like
Treating the patient with respect	Introduce yourself, address them as Mr. or Ms. Lee, explain before you act, ask permission for positioning, acknowledge discomfort, and thank them.

Domain element	What it looks like
Sharing information effectively	Explain in plain language, in a logical order, at a reasonable pace, without jargon, checking understanding.
Receiving information effectively	Let the patient finish. Do not interrupt. Reflect back what they said ("So the flashing started three days ago and the shadow came yesterday, is that right?"). Ask open questions before closed ones.
Collaborating with the patient	Offer options where they exist, ask about preferences and constraints, and acknowledge the practical impact of what you are proposing.
Collaborating with other professionals	Say out loud who you are involving and why: "I am going to telephone the retina service now and speak to them directly, and I will write to your family doctor as well."
Managing your own manner	Examiners, proctors, and SPs are trained to be neutral and to show little emotion . Do not read that as disapproval, and do not let it flatten your own manner. Stay warm and steady.

WHAT TO DO WHEN A STATION GOES BADLY

1. **Do not abandon the remaining items.** Each item is scored independently. A missed quadrant does not affect the next one.
2. **Do not chase the lost item** with four attempts. Two attempts, then move on.
3. **Do not announce a skill repeat** unless you are early and the error was fundamental.
4. **Keep narrating.** Description items are still available even if the technique item was lost.
5. **Reset your manner.** The SP scores communication in this station and knows nothing about the previous one.
6. **File an incident report in the break** if something outside your control contributed.
7. **Then let it go.** There are eleven other stations, one of the ten encounters does not count at all, and the pass decision is made on the whole session.

RETRIEVAL PRACTICE

1. Where is the slit lamp camera, and what does that mean for you?
2. What are the only two things an examiner will say about a view?
3. What does a confirmed view *not* tell you? Give three things.
4. What happens if you ask about a view after finishing the skill?
5. How many attempts per item, and what happens after the last one?
6. What are the consequences of a stop?
7. What is scored if you repeat an item?
8. What happens to your scores if you announce a skill repeat?
9. Give the two cautions the guide makes about repeating a skill.
10. Can a stopped skill be repeated?
11. What must you do if the SP or examiner intervenes?
12. Name five ways to break hygiene protocol, and the commonest one.
13. What happens if you say "I am finished with this station"?
14. Where does additional time appear if it is granted?
15. May you look at the monitor in the room?
16. Give four behaviours that earn the communication and professionalism domain.

KEY

1. On the **left ocular**; you are scored on what the camera captures there, so the image must be fully seen through the left ocular.

2. **"I have a view"** or **"I do not have a view."**

3. The **quality** of the view, whether it **meets the item criteria**, and whether it is of the **correct structure, angle, or quadrant**.

4. You are told **"I can only comment on a view while it is being performed."**
5. **Four.** On unsuccessfully completing the fourth you are stopped and told to move on.
6. All **remaining items in that skill are scored "no"**, and the skill **cannot be repeated**.
7. The **last intentional performance** of that item.
8. **All previously recorded scores for that skill are erased**, and you are evaluated as if performing it for the first time, so every item must be redone.
9. Candidates who repeat an entire skill for one item frequently **run out of time**, and candidates repeating a skill often **overlook other items and score lower than they had already earned**.
10. **No.**
11. **Take corrective action before proceeding**, and do so explicitly.
12. Touching your face or hair after washing (the commonest), not washing at all, failing to state you would disinfect the instruments, touching the eye or lashes with a bottle tip, contaminating a lens, not covering an open wound, and reusing an applicator.
13. Scoring ends, the examiner leaves the room, the SP remains and begins preparing the room, and **no further items are scored**.
14. In the **top left corner of your device**.
15. **No, at any time.** Attempting to do so may be treated as improper conduct.
16. Introducing yourself and using the patient's name; explaining before acting and warning about light and touch; listening without interrupting and reflecting information back; offering options and acknowledging impact; and stating who you are involving in their care and why.

CHAPTER RECAP

1. Ask about views early, and know that confirmation means only that something is visible.
2. Nothing may obstruct the left ocular or the central BIO camera, and you may never look at the monitor.
3. Four attempts then a stop, and a stop scores every remaining item in that skill as "no".
4. Two attempts is the practical limit; then move on.
5. Repeat items freely; repeat skills almost never, because announcing it erases everything.
6. A stopped skill cannot be repeated.
7. Take visible corrective action after any intervention.
8. Hygiene is scored: wash during observation, then do not touch your face or hair.
9. Warn before every touch and every bright light, respect personal space, and never touch below the shoulder.
10. A bad station is one station. Keep narrating, reset your manner, and move on.

Rehearsal plans and timed drills

Everything here is done out loud and on a clock

FIVE RULES FOR EVERY PRACTICE SESSION

1. **Out loud, always.** Never rehearse a skill silently. The verbalisation is the score.
2. **On a timer, always.** Fifteen minutes, from the first practice session onward.
3. **Against the checklist.** Use the appendix forms and have a partner tick items as you perform them, then count the misses.
4. **On unfamiliar people.** Rotate partners. Practise on people whose fundus you have not memorised.
5. **Log every miss.** Keep one page per skill listing the item numbers you lose, and target those specifically in the next session.

Eight-week plan

THE FULL PREPARATION ARC

Week	Focus	Sessions
1	Learn the forms	Read the current candidate guide, blueprint, and both evaluation forms end to end. Transcribe the evaluation forms into your own checklist. Watch the NBEO orientation video and complete the software tutorial. Write out your own scripts for the five skills, in your own words, and read them aloud until they are fluent.
2	Biomicroscopy, untimed then timed	Three sessions. First untimed, narrating every one of items 2 to 25 while a partner ticks. Then two timed runs at 6 minutes . Drill upper lid eversion twenty times until it takes eight seconds. Drill Van Herick and iris transillumination separately, ten repetitions each.
3	Gonioscopy and appplanation	Three sessions. Gonioscopy: mirror-to-quadrant naming drill on paper, then twenty lens insertions and removals, then four full timed runs at 4.5 minutes covering all four quadrants with structure naming. Appplanation: ten full timed runs at 4 minutes , including stating the pressure with the time and educating the patient.
4	Full anterior station	Four complete 15-minute anterior stations with a partner scoring all 59 items. Target: fewer than five misses by the end of the week. Include a deliberate "lost view" drill where your partner tells you they have no view, so you practise troubleshooting under time.
5	BIO	Four sessions. First, quadrant geography drill: for each of the eight quadrants, state where you stand and where the patient looks, on paper, until instant. Then eight-quadrant runs at 40 seconds per quadrant with a partner calling the time. Specifically drill finding and naming a vortex vein ampulla in every quadrant.
6	Dilated biomicroscopy and fundus lens	Four sessions. Drill lens layer scanning with verbalisation, retroillumination at maximum illumination with a moving light source, and fundus lens positioning. Then full runs of items 22 to 43 at 7 minutes , insisting on both cup-to-disc numbers, the nasal arcades, and a separate fovea description.

Week	Focus	Sessions
7	Full posterior station, and encounter drills begin	Three complete 15-minute posterior stations scoring all 44 items. In parallel, begin encounter station drills : three per session, using the case categories in Chapter 4, with a partner playing the SP from a written script.
8	Full mock session and polish	One full twelve-station mock if you can arrange it, or at minimum a six-station half-session, with real timing and real breaks. Then targeted repair of your top five recurring misses. Two days before, stop practising skills and review only your scripts, the lexicon, and the plan templates.

Four-week plan

Week	Content
1	Read the guide and both forms. Build your checklists. Write and memorise your five scripts. Three biomicroscopy runs and three applanation runs, timed.
2	Gonioscopy geography and technique to fluency. Two full timed anterior stations scored against the checklist. Begin BIO quadrant geography drills.
3	BIO eight-quadrant runs at 40 seconds each. Fundus lens runs. Two full timed posterior stations scored. Six encounter drills, focusing on the four-minute history and the four-minute note.
4	One anterior and one posterior full station on each of three days, plus six encounter drills. Final two days: scripts, lexicon, and plan templates only.

One-week plan, for when time has run out

Day	Content
1	Read both evaluation forms aloud, item by item, and write your script for each of the five skills. Read the candidate guide's rules on views, repeats, and stops.
2	One timed biomicroscopy run and one timed applanation run, scored. Drill upper lid eversion, Van Herick, and iris transillumination twenty times each.
3	Gonioscopy: mirror-to-quadrant drill, then three timed runs naming quadrant, most posterior structure, and iris approach.
4	One full timed anterior station, scored, then repair the misses immediately in a second run.
5	BIO quadrant geography drill, then two eight-quadrant timed runs. Fundus lens run covering items 29 to 43.
6	One full timed posterior station, scored. Then four encounter drills: history in four minutes and a complete typed plan in four minutes.
7	Rest. Read your scripts, the lexicon, and the plan templates once. Sleep.

Individual drills

TEN DRILLS, WITH THE TARGET

Drill	Target	Method
Upper lid eversion	8 seconds, 20 for 20	With a cotton-tipped applicator, on five different partners. Instruct, evert, sweep medial to lateral, describe, release, and have them blink. Count failures.

Drill	Target	Method
Whole-cornea sweep	30 seconds each	Parallelepiped limbus to limbus in overlapping passes, then optic section limbus to limbus, narrating both.
Van Herick	15 seconds	Beam at 60 degrees at the temporal limbus, state the shadow-to-thickness ratio and the grade. Ten repetitions on five people, comparing your grades against a partner's.
Cells and flare	20 seconds	Darken the room, small bright conical beam, focus in the aqueous, state grade zero cells and grade zero flare using SUN language.
Iris transillumination	20 seconds	Beam through the pupil, moved across it, covering the whole iris; narrate the technique and the finding as separate items.
Gonioscopy quadrant naming	Instant	Paper drill: partner says "temporal mirror", you say "nasal angle". Then reverse. Fifty repetitions. Then on the lens.
Applanation mire endpoint	45 seconds per eye	From prism contact to a stated reading with the time. Have your partner deliberately over- or under-fluoresce so you learn to recognise and correct mire width.
BIO quadrant geography	Instant	Partner names a quadrant, you state where you stand and where the patient looks, and then do it. Eight quadrants, ten cycles.
Vortex ampulla hunt	8 for 8	In each quadrant, push out until an ampulla is visible, hold it, and name it. This single drill is the difference between eight scored peripheral views and eight unscored ones.
Four-minute typed plan	4 minutes	Partner reads a case from Chapter 4. You type a complete plan on a laptop: medical, non-medical, referral with specialty and timeframe, additional testing with indications, patient education with mechanism and warning signs, and a single RTC interval. Score it against the template.

THE FOUR-MINUTE HISTORY DRILL, WHICH MOST CANDIDATES SKIP

Have a partner sit opposite you with a written script for one of the cases in Chapter 4. They answer only what is on the script, and say "**I do not have that information**" for anything else, exactly as the SP will. Set a timer for **4 minutes**.

Score yourself on coverage: did you obtain the HPI in full, past ocular history including correction and compliance, medical history with medications and allergies, surgical history, family ocular and medical history, and social history including occupation, driving, smoking, and eye protection? Six categories in four minutes.

Do this drill **fifteen times** across your preparation. It builds the sequence into muscle memory, and the sequence is what protects you when the case is unfamiliar and you feel the time pressure.

The week before, and the day itself

LOGISTICS AND MINDSET

When	Do
One week before	Download and re-read the current candidate guide, blueprint, and both evaluation forms and reconcile them with your checklists, because item numbering changes between administrations. Confirm your registration date and time. Book travel and accommodation close to the NCCTO. Complete the software tutorial again.
Two days before	Stop practising skills. Read your scripts, the lexicon, and the plan templates. Pack: lab coat, professional clothes and closed shoes, two forms of identification (one government photo ID), your OE TRACKER number and NBEO password, an eye patch if you use one, permitted small items, and a labelled drink and snack for the hallway bins.
Night before	Sleep. Do not review anything after the evening. Set two alarms. Plan to arrive at the building no earlier than 15 minutes before your listed time.
Morning	Eat properly. Tie hair back. Wear your glasses if you wore them in your check-in photograph. Use the restroom before the session starts, because no extra time is given.
During orientation	Handle the actual laptop, BIO headset, and slit lamp. Adjust the BIO headset to your own pantoscopic tilt and ocular separation and remember the settings. Note where supplies are kept. Ask staff any equipment question now, because instructional questions will not be answered later.
Between stations	Sit at your designated desk for the <i>next</i> room number. File an incident report if needed. Drink water; eat if you need to, use the restroom. Do not replay the last station. Take three slow breaths and reset.
After cycle 8	Use the 24-minute break properly: eat something, move, and rehydrate. Fatigue in the last four stations is a real and preventable cause of lost items.
At the end	File any final incident report at your desk and remain seated until dismissed. Reports are not accepted once you leave the hallway.

THE TEN SENTENCES TO HAVE MEMORISED VERBATIM

1. "I am candidate [OE TRACKER number]. Which eye am I examining today?"
2. "Before I begin, I would properly disinfect all ocular instruments prior to using them."
3. "Please look all the way down and keep looking down. I am going to turn your upper eyelid out for a moment."
4. "I am performing a Van Herick assessment at the temporal limbus; the shadow depth is greater than one half of the corneal section thickness, grade 4."
5. "I am transilluminating the entire iris by directing the beam through the pupil."
6. "I am viewing the inferior angle through the superior mirror. The most posterior visible structure is the ciliary body band, the iris approach is regular and flat, Shaffer grade 4."
7. "The intraocular pressure of the right eye by Goldmann applanation is 16 millimetres of mercury at [time]."
8. "Your eye pressure today was 16, which is in the normal range. Pressure matters because pressure that is too high over time can damage the nerve at the back of the eye."
9. "I am viewing the superior-temporal peripheral retina, from the equator to the ora serrata, with a vortex vein ampulla in view. It is flat and intact with no holes, tears, or lattice degeneration."
10. "Horizontal cup-to-disc ratio 0.3. Vertical cup-to-disc ratio 0.3."

Plus two for the encounter stations: **"Please give me a moment to review some information on my device"** and **"Thank you, Ms. Lee, you may check out."**

CHAPTER RECAP

1. Out loud, on a clock, against the checklist, on unfamiliar people, with a log of misses.
2. Eight weeks if you have them, four if you do not, one if that is all there is.
3. Upper lid eversion, Van Herick, iris transillumination, and the vortex ampulla hunt are the four highest-yield individual drills.
4. Forty seconds per BIO quadrant is the pacing that makes the posterior station work.
5. Practise typing a complete plan in four minutes, fifteen times.
6. Practise a six-category history in four minutes, fifteen times.
7. Download the current forms in the week before, because item numbering changes.
8. Use orientation time to set up the actual BIO headset and slit lamp.
9. Use the 24-minute break to eat and reset, because fatigue costs items in the last four stations.
10. Memorise the ten sentences verbatim, so they arrive without effort under pressure.

APPENDIX

Printable scored-item checklists

These checklists reproduce the item structure of the official **Anterior Segment Skills Evaluation Form** and **Posterior Segment Skills Evaluation Form** for the August 2026 to May 2027 administration, so that a study partner can score you item by item during a timed mock station. Tick each item only if you both *performed* and, where the item requires it, *verbalised* it.

Verify against the current forms on the NBEO website before your exam date, because item numbering and wording change between administrations.

Anterior segment skills station

General item

1. Stated they would properly **disinfect ocular instruments** prior to using them.

Skill 1 • Biomicroscopy (items 2 to 25)

1. Explained the purpose of the procedure to the patient.
2. Properly examined the **entire** upper and lower lids and lashes.
3. Accurately described the upper and lower lids and lashes.
4. Properly examined the **entire** upper and lower lid margins and the **lower puncta**.
5. Accurately described the upper and lower lid margins and the lower puncta.
6. Properly examined the **entire inferior palpebral conjunctiva** using lower lid eversion.
7. Accurately described the inferior palpebral conjunctiva.
8. Properly examined the **entire superior palpebral conjunctiva** using upper lid eversion.
9. Accurately described the superior palpebral conjunctiva.
10. Properly examined the **entire** bulbar conjunctiva and sclera.
11. Accurately described the bulbar conjunctiva and sclera.
12. Properly examined the **entire cornea using a parallelepiped**.
13. Properly examined the **entire cornea using an optic section**.
14. Accurately described the cornea (**epithelium, stroma, endothelium**).
15. Properly assessed the **temporal anterior chamber angle using the Van Herick technique**.
16. Accurately described the temporal anterior chamber angle findings using the Van Herick technique.
17. Properly examined the anterior chamber for **cell and flare**.
18. Accurately described the anterior chamber cell and flare.
19. Properly examined the **entire iris**.
20. Accurately described the iris.

21. Properly performed **transillumination of the entire iris**.
22. Accurately described the iris transillumination.
23. Examined the ocular tissues in a **smooth, logical sequence**.
24. Properly **instructed the patient** throughout the procedure.

Skill 2 • Four-mirror gonioscopy (items 26 to 41)

1. Explained the purpose of the procedure **including the drop to be used**.
2. Properly prepared the gonioscopy lens and installed **anaesthetic into the appropriate eye**.
3. Provided proper instructions to the patient.
4. Inserted and utilised the gonioscopy lens in a **safe and efficient** manner.
5. Obtained and maintained a clear image of the **entire inferior quadrant**.
6. Accurately identified the inferior quadrant, the **most posterior visible angle structure**, and the **iris approach**.
7. Obtained and maintained a clear image of the **entire nasal quadrant**.
8. Accurately identified the nasal quadrant and the most posterior visible angle structure.
9. Obtained and maintained a clear image of the **entire superior quadrant**.
10. Accurately identified the superior quadrant and the most posterior visible angle structure.
11. Obtained and maintained a clear image of the **entire temporal quadrant**.
12. Accurately identified the temporal quadrant and the most posterior visible angle structure.
13. Maintained **control of the gonioscopy lens at all times**.
14. Removed the gonioscopy lens in a **safe and efficient** manner.
15. Assessed and accurately described **corneal integrity without staining** at the conclusion.
16. Performed the procedure in a **smooth and logical sequence**.

Skill 3 • Goldmann applanation tonometry (items 42 to 58)

1. Explained the purpose of the procedure **including the drop to be used**.
2. Instilled the appropriate **anaesthetic and fluorescein combination properly in one eye**.
3. Adjusted the biomicroscope **illumination system** properly.
4. Assessed and described the cornea for **staining prior to tonometry**.
5. Assessed and described the **quality of the tear film prior to tonometry**.
6. Set the pressure on the probe at an appropriate level.
7. Properly positioned the tonometer and aligned the probe.
8. Provided proper instructions to the patient, **including a fixation target**.
9. Properly applanated the cornea in a **safe and efficient** manner.
10. Adjusted the biomicroscope so the fluorescein semicircular patterns are **symmetrical and centred**.

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|-----|---|
| 11. | Ensured the tonometer mires are of the appropriate width . |
| 12. | Properly aligned the tonometer mires . |
| 13. | Safely retracted the tonometer. |
| 14. | Accurately stated the obtained finding with the time . |
| 15. | Assessed and accurately described the cornea using the blue filter following tonometry . |
| 16. | Performed the procedure in a smooth and logical sequence . |
| 17. | Educated and explained the patient's IOP. |
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General item

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| 1. | Maintained proper hygiene throughout the station. |
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Scoring the mock: 59 items. Record the number ticked, the elapsed time at which each skill was completed, and the item numbers missed. Items marked with a crimson box in this checklist are the ones most commonly lost; check those first when reviewing.

Posterior segment skills station

Skill 4 • Binocular indirect ophthalmoscopy (items 1 to 21)

1. Explained the purpose of the procedure to the patient.
2. Properly examined and obtained a clear image, **filling the condensing lens as much as possible without the arcades in view**, of the **superior** peripheral retina.
3. Accurately identified the quadrant and provided an assessment of the superior peripheral retina.
4. Properly examined and obtained a clear image of the **superior-nasal** peripheral retina.
5. Accurately identified the quadrant and provided an assessment of the superior-nasal peripheral retina.
6. Properly examined and obtained a clear image of the **nasal** peripheral retina.
7. Accurately identified the quadrant and provided an assessment of the nasal peripheral retina.
8. Properly examined and obtained a clear image of the **inferior-nasal** peripheral retina.
9. Accurately identified the quadrant and provided an assessment of the inferior-nasal peripheral retina.
10. Properly examined and obtained a clear image of the **inferior** peripheral retina.
11. Accurately identified the quadrant and provided an assessment of the inferior peripheral retina.
12. Properly examined and obtained a clear image of the **inferior-temporal** peripheral retina.
13. Accurately identified the quadrant and provided an assessment of the inferior-temporal peripheral retina.
14. Properly examined and obtained a clear image of the **temporal** peripheral retina.
15. Accurately identified the quadrant and provided an assessment of the temporal peripheral retina.
16. Properly examined and obtained a clear image of the **superior-temporal** peripheral retina.
17. Accurately identified the quadrant and provided an assessment of the superior-temporal peripheral retina.
18. Properly examined and obtained a clear image, filling the condensing lens as much as possible, of the **posterior pole**.
19. Provided an accurate assessment of the posterior pole.
20. Performed the skill in a **safe manner for the patient**.
21. Performed the procedure in a **smooth and logical sequence**.

Skill 5 • Dilated biomicroscopy and non-contact fundus lens evaluation (items 22 to 43)

1. Explained the purpose of the procedure to the patient.
2. Properly **scanned all layers of the crystalline lens**.
3. Accurately described the crystalline lens.

4. Properly examined the crystalline lens using **retroillumination (maximum illumination for this item only, moving the light source)**.
5. Accurately described the crystalline lens using retroillumination.
6. Properly examined the **anterior vitreous**.
7. Accurately described the anterior vitreous.
8. Provided appropriate **instructions for fixation**.
9. Properly positioned the non-contact fundus lens (**alignment, centration, distance**).
10. Properly examined the **posterior vitreous**.
11. Accurately described the posterior vitreous.
12. Properly examined the **optic nerve**.
13. Accurately described the optic nerve (**colour, rim integrity, contour, PPA**).
14. Accurately stated the **horizontal cup-to-disc ratio**.
15. Accurately stated the **vertical cup-to-disc ratio**.
16. Properly examined and obtained a clear image of the **inferior and superior temporal arcades**.
17. Accurately identified and provided an assessment of the inferior and superior temporal arcades.
18. Properly examined and obtained a clear image of the **inferior and superior nasal arcades**.
19. Accurately identified and provided an assessment of the inferior and superior nasal arcades.
20. Properly examined **both the fovea and the macula**.
21. Accurately described **both the fovea and the macula**.
22. Performed the procedure in a **smooth and logical sequence**.

General item

1. Maintained **proper hygiene** throughout the station.

Scoring the mock: 44 items. Record the number ticked, the elapsed time at the end of BIO, and the item numbers missed.

Patient encounter station self-audit

There is no published item list for the encounter stations, because they are scored through the note and through the SP's communication matrix. This audit maps onto the five competency domains and can be used to score a mock encounter.

Skills domain: case history

1. Introduced self by OE TRACKER number and addressed the patient as Mr. or Ms. Lee.
2. Obtained the full HPI: location, onset, duration, severity, frequency, modifying factors, associated symptoms, prior episodes, further description.
3. Obtained past ocular history: conditions, duration, treatment, surgery, injury, last examination.
4. Obtained correction history: type, age of prescription, wear pattern, compliance.
5. Obtained medical history: conditions, duration, control, last physical examination.
6. Obtained a complete medication list, including over-the-counter agents, supplements, and drops.
7. Obtained allergies **and the reaction**.
8. Obtained surgical history.
9. Obtained family ocular **and** family medical history.
10. Obtained social history: occupation, hobbies, driving, smoking, alcohol, eye protection.
11. Asked at least one follow-up question **after** reviewing the clinical data.
12. Completed the history within 4 minutes 30 seconds.

Clinical assessment and interpretation

1. Identified every abnormal finding in the clinical data.
2. Ordered specific ancillary tests, each with an indication, before submitting once.
3. Correctly interpreted the ancillary results provided, including recognising normal results.
4. Selected the **most significant** diagnosis of the visit from the diagnosis list.
5. Selected the correct **location** (OD, OS, OU, or N/A).

Management and documentation

1. Documented medical therapy with drug, concentration, frequency, duration, and instructions.
2. Documented non-medical therapy where applicable.
3. Documented any referral with **specialist, reason, and timeframe**.
4. Documented additional testing with rule-in or rule-out indications.
5. Documented a single **return-to-clinic interval**, no range and no calendar date.

Patient education

1. Named the diagnosis and then explained it in plain language.

2. Explained the cause or mechanism.
3. Explained the treatment, how to use it, and its expected effect and side effects.
4. Explained warning signs requiring earlier return.
5. Stated when the patient should come back.
6. Checked the patient's understanding.
7. Documented the education in the note in patient-centred language.

Communication and professionalism

1. Used no jargon after naming the diagnosis.
2. Did not interrupt, and reflected information back to confirm it.
3. Spoke to the patient rather than the screen, and signalled device use appropriately.
4. Offered options and acknowledged the practical impact of the plan.
5. Stated which other professionals would be involved and why.
6. Dismissed the SP explicitly, and only after all questions and all education were complete.
7. Began typing the note by **minute 11**.

FINAL CHECKLIST BEFORE YOU TRAVEL

1. Have you downloaded and read the **current** candidate guide, blueprint, and both evaluation forms, and reconciled them with these checklists?
2. Have you completed the software tutorial and watched the orientation video?
3. Can you recite the ten verbatim sentences without thinking?
4. Have you scored at least three full anterior and three full posterior mock stations against these forms?
5. Can you complete a six-category history in four minutes and a complete typed plan in four minutes?
6. Do you know where you stand and where the patient looks for all eight BIO quadrants, instantly?
7. Can you find and name a vortex vein ampulla in every quadrant?
8. Do you have your lab coat, two forms of identification, your OE TRACKER number, and your NBEO password?
9. Do you know your reporting date, time, and address, and have you allowed for travel?
10. Have you planned to arrive no earlier than fifteen minutes before your listed time, with nothing that counts as luggage?